

Reviewed Preprint

v1 • September 16, 2025

Not revised

Reviewed Preprint

v2 • March 2, 2026

Revised by authors

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Competing interests:

No competing interests declared

Funding:

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Reviewing editor: Jungsan Sohn,
Johns Hopkins University School of
Medicine, United States

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Adaptor protein supersaturation drives innate immune signaling and cell fate

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eLife Assessment

This **important** study investigates the self-assembly activity of all 109 human death-fold domains. The data collected using advanced microscopy and distributed amphifluoric FRET-based flow cytometry methods are **compelling** to support the "phase change battery" model that explains how signal amplification can occur without ATP consumption. This paper provides new insight into the thermodynamic control of protein phase behaviors within cells and will be of interest to those studying a variety of biological pathways involved in inflammatory responses and various forms of cell death.

<https://doi.org/10.7554/eLife.107962.2.sa3>

Abstract

How minute pathogenic signals trigger decisive immune responses is a fundamental question in biology. Classical signaling often relies on ATP-driven enzymatic cascades, but innate immunity frequently employs death fold domain (DFD) self-assembly. The energetic basis of this assembly is unknown. Here, we show that specific DFDs function as energy reservoirs through metastable supersaturation. Characterizing all 109 human DFDs, we identified sequence-encoded nucleation barriers specifically in the central adaptors of inflammatory signalosomes, allowing them to accumulate far above their saturation concentration while remaining soluble and poised for activation. We demonstrate that the inflammasome adaptor ASC is constitutively supersaturated *in vivo*, retaining energy that powers on-demand cell death. Swapping a non-supersaturable DFD in the apoptosome with a supersaturable one sensitized cells to sublethal stimuli. Mapping all DFD nucleating interactions revealed that supersaturated adaptors are triggered to polymerize specifically by other DFDs in their respective pathways, limiting potentially deleterious crosstalk. Across human cell types, adaptor supersaturation strongly correlates with cell turnover, implicating this thermodynamic principle in the trade-off between immunity and longevity. Profiling homologues from fish and sponge, we find nucleation barriers to be conserved across metazoa. These findings reveal DFD adaptors as biological phase change materials in thermal batteries to power cellular life-or-death decisions on demand.

Main

Innate immune signaling transduces small signals to robust responses such as programmed cell death and/or inflammation. Signaling occurs when danger- or pathogen-associated molecular patterns (D/PAMPs) activate cognate receptor proteins that then activate effector proteins such as caspases, typically via one or more intermediary proteins known as adaptors.

While the binding of a D/PAMP to receptors initiates signaling, this initial interaction releases insufficient energy to directly change cell state. That tiny signal must be amplified in an energy consuming process that is fundamental to understanding the architecture and evolution of these

critical signaling networks.^{1–3} Innate immune signaling networks from bacteria to humans amplify signaling through interactions between nonenzymatic death fold domains (DFDs), which comprise the caspase recruitment domain (CARD), death domain (DD), death effector domain (DED), and pyrin domain (PYD) subfamilies.^{4–8} However, the energetic basis for DFD function has been the subject of considerable speculation but remains largely unresolved (Figure 1A [↗](#)).

DFDs commonly form paracrystalline polymers that can template their own growth when free subunits exceed their saturation concentration (C_{sat}). Polymers are functionally initiated by D/PAMP-bound receptor oligomerization. To preclude signaling through spontaneous nucleation, DFDs are presumed to be effectively subsaturated prior to activation, through a combination of low basal expression, subcellular compartmentalization, post-translational modifications, and regulatory interactions.^{4,9–11} We previously discovered, however, that the DFD-containing adaptor Bcl10 exhibits an intrinsic (sequence-encoded) nucleation barrier that is large enough to support persistent deep supersaturation in vivo, allowing it to amplify signaling independently of orthogonal input.¹²

The increased ordering of DFD subunits in polymers relative to those in solution implies that polymerization releases heat. Here, we propose that DFD-containing signalosomes broadly function as a form of thermal battery – an on-demand energy reservoir that can be discharged via latent heat release through a specific external input.¹³ Energy storage is accomplished via metastable supersaturation by a phase change material. A familiar example of a phase change material is the sodium acetate solution inside reusable hand warmers, which releases heat when the metal disk inside is pressed to nucleate crystalline sodium acetate trihydrate. Supersaturation is a state where a solute's concentration exceeds C_{sat} yet remains in solution due to a structurally determined nucleation barrier. In the context of innate immune signalosomes, at least one DFD would function as the phase change material by constitutively exceeding its C_{sat} , allowing the signalosome to assemble immediately upon D/PAMP-triggered nucleation. Although heat will necessarily be released during the phase transition, it dissipates far too quickly to influence downstream signaling.¹⁴ This mechanism would allow cells to respond to D/PAMPs decisively and independently of metabolism, which is frequently compromised during infection.¹⁵ For DFDs to function in this manner, their endogenous concentration would need to greatly exceed their respective C_{sat} values, while they nevertheless remain soluble over timescales spanning the window of vulnerability to infection, i.e. the full lifetimes of cells. Cells would therefore be continuously susceptible to spontaneous death and inflammation through stochastic nucleation events, imposing a fundamental tradeoff between immunity and longevity.

We here use a combination of biophysical, bioinformatic, and cytological approaches to investigate the capacity and functional relevance of supersaturation by human DFDs. Our results collectively uncover the energetic basis for signal amplification by DFDs and in turn a thermodynamic drive to die.

A select subset of DFDs have intrinsic nucleation barriers enabling persistent supersaturation

To systematically survey the ability of DFDs to supersaturate, we compiled an exhaustive set of 109 structurally independent human DFDs (Table S1 and Figure S1A-B) and then characterized their tendency to spontaneously self-assemble in near-physiological conditions while minimizing interference from other proteins. For this purpose we used distributed amphifluoric FRET (DAmFRET) in an orthogonal eukaryotic host that completely lacks endogenous DFDs and their associated regulatory machinery – *Saccharomyces cerevisiae*. DAmFRET produces a snapshot of the population-level distribution of single-cell measurements of ratiometric FRET (“AmFRET”) between two fluorescent forms of the same protein species.¹⁶ The data revealed a diversity of behaviours (Figure 1B-C [↗](#), S1C-D, and Table S1), ranging from no self-association to self-association in all cells at all concentrations. Importantly, the results closely agree with observations of individual DFDs in the literature. Considering the adaptor FADD as an example, its DD was monomeric whereas its DED assembled robustly, consistent with prior observations that

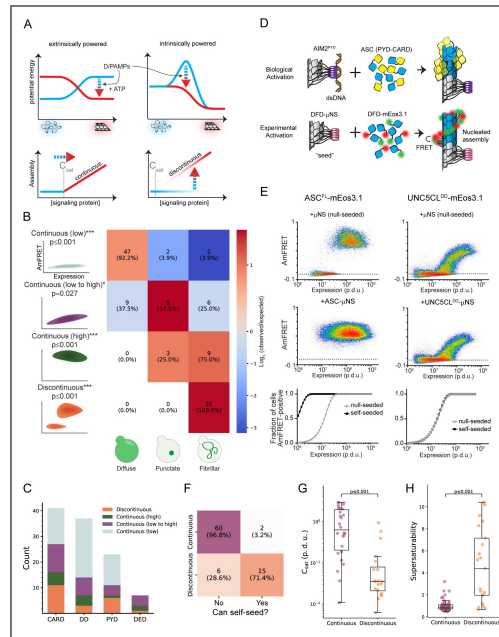


Figure 1. A select subset of DFDs have intrinsic nucleation barriers enabling persistent supersaturation.

A. Schematic diagram illustrating two models for signal amplification through protein selfassembly. **Top left:** Extrinsic model, where D/PAMP-binding coupled with nucleotide hydrolysis stabilizes active assemblies (red glow) relative to solute precursors (blue glow). This model is exemplified by localized actin polymerization downstream of many cell surface receptors,¹¹⁷ but could also occur indirectly by, for example, phosphorylation-mediated release of solubilizing factors.

Bottom left: DFDs that function in this way will assemble promptly and monotonously above their saturation concentration (C_{sat}). **Top right:** Intrinsic model, where the protein is supersaturated at rest but prevented from assembling by a sequence-encoded nucleation barrier. D/PAMPbinding eliminates the barrier, releasing the energy of supersaturation to drive assembly. The models are not mutually exclusive. **Bottom right:** DFDs that function in this way will remain soluble above C_{sat} until stochastic nucleation, creating a discontinuous relationship of assembly to concentration across a population of cells. B. Illustration showing how the concentration-dependence of self-assembly as classified by DAMFRET relates to the subcellular morphology of self-assemblies classified by highthroughput confocal microscopy. “Continuous” and “discontinuous” classifications describe the relationship of self-assembly (AmFRET, y-axis) to expression level (x-axis) for each DFD. Discontinuous DFDs exhibit a range of concentrations where selfassembly occurs stochastically, indicating an intrinsic nucleation barrier. The four instances of visible assemblies despite no AmFRET-positive cells are presumed to result from those DFDs partitioning with other cellular components or endogenous condensates wherein they remain too dilute to FRET. Cells in the matrix are colored according to the log₂ ratio of observed to expected frequencies. The indicated p-values (adjusted for multiple hypotheses) were obtained with an exact multinomial test using the total frequencies of each morphology. C. Distribution of DAMFRET classifications across the four subfamilies of DFDs. D. Schematic diagram of our experimental design to assess the ability of each DFD to seed itself. Top: Biological activation of an exemplary signalosome -- the AIM2 inflammasome -- occurs when the receptor AIM2 oligomerizes on the multivalent PAMP, dsDNA, and then templates the assembly of the adaptor protein, ASC. Bottom: Experimental paradigm to test for supersaturation mimics biological activation, by expressing each DFD in trans with the same DFD expressed as a fusion to μ NS, a modular self-condensing protein. AmFRET-positivity will only occur if the μ NS fusion templates subsequent self-assembly by the non μ NS-fused DFD. E. Representative DAMFRET data contrasting two self-assembling DFDs -- one that is supersaturable (left) and the other that is not (right). The plot for the supersaturated protein exhibits a discontinuous distribution of AmFRET across the expression range (top and bottom). The discontinuity is eliminated, with all cells moving to the AmFRETpositive population, by expressing the protein in the presence of genetically encoded seeds (middle). The dashed horizontal lines approximate the mean AmFRET value for monomeric mEos3. Procedure defined units (p.d.u.). F. Contingency table showing that discontinuous DFDs tend to be self-seedable. Each DFD was co-expressed with an orthogonally fluorescent μ NS-fused version of the same DFD. Fisher’s exact test revealed an association between continuity and self-seedability ($p < 0.001$). G. Boxplot comparing the C_{sat} values (as approximated by $C50_{seeded}$) of continuous and discontinuous DFDs. Discontinuous DFDs have significantly lower C_{sat} , indicating greater stability of the assemblies. Mann-Whitney U = 457, $n_{continuous} = 26$, $n_{discontinuous} = 20$ ($p < 0.001$). H. Boxplot comparing supersaturability, represented as the fold change reduction in $C50$ by seeding ($C50_{stochastic} - C50_{seeded}$), of continuous and discontinuous DFDs. The $C50$ values were more strongly reduced by seeding for discontinuous DFDs than for continuous DFDs. Mann-Whitney U = 164, $n_{continuous} = 58$, $n_{discontinuous} = 21$ ($p < 0.001$). See also Figures S1, S2, S5 and Table S1.

the DD forms exclusively hetero-oligomers while the DED forms homopolymers.^{17–19} Twenty-one DFDs transitioned from no to high AmFRET in a discontinuous manner, the signature of a large intrinsic nucleation barrier.^{12,16,20}

Nucleation barriers increase with the entropic cost of assembly.^{16,21} Consequently, assemblies with large barriers tend to be more ordered than those without. Ordered assembly by DFDs occurs as a two-dimensional array twisted into a one-dimensional polymer^{22–24} that can manifest as microscopically visible filaments in cells.^{12,23,25} To evaluate the expected relationship between nucleation barriers and ordered assembly, we used high-throughput confocal microscopy to examine the subcellular distribution of each protein. We observed subcellular assemblies for most of the DFDs that populated AmFRET-positive states, but not those with entirely low AmFRET (Figure 1B [↗](#) and S1E). As predicted, the assemblies of DFDs that had transitioned discontinuously had fibrillar morphologies, whereas those that had transitioned continuously (low to high DAMFRET) instead mostly formed spherical or amorphous puncta (Figure 1B [↗](#), S1F–H).

For discontinuous DFDs, the soluble protomers in the low FRET cells are hypothetically poised to assemble; they just lack a structural template to get them started. We tested this hypothesis by co-expressing genetically encoded “seeds” (Figure 1D–E [↗](#)),¹² and found that the seeds caused cells to switch to the high AmFRET population for most discontinuous DFDs but not continuous DFDs (Figure 1F [↗](#), S1I, Table S1). We further demonstrated that assembly is more consistent with native DFD subunit interfaces than amyloid-like misfolding (Supplemental Information, Figure S2A–B). This distinction is important because amyloid is the only other form of assembly that has been experimentally shown to be capable of forming a discontinuous DAMFRET profile.^{16,26} Altogether, these data confirm that a subset of DFDs can supersaturate in a soluble form to power subsequent switch-like assembly.

The magnitude of supersaturation in vivo is determined by the ratio of a protein’s total concentration to its C_{sat} , which reflects the strength of interactions between subunits in its assembled structure.²⁷ Total concentration and C_{sat} can evolve independently by gene regulation and DFD sequence, respectively. Consequently, if the function of discontinuous DFDs involves supersaturation, we would expect evolution to have lowered their C_{sat} while raising their expression relative to continuous DFDs. To test these predictions, we first analyzed the relationship of nucleation barriers to C_{sat} as determined by each protein’s transition concentration in the presence of seed. As expected, discontinuous DFDs exhibited lower C_{sat} values than continuous DFDs (Figure 1G [↗](#), Table S1), and in the absence of seed achieved soluble concentrations that exceeded them by at least four-fold on average (Figure 1H [↗](#)).

Nucleation barriers are restricted to innate immune signalosome adaptors

We next evaluated the relationship of nucleation barriers to DFD concentrations in vivo. Using published proteomic datasets,^{28–30} we found that proteins with discontinuous DFDs tend to be expressed more abundantly than those with continuous DFDs, both at the cell type and tissue levels (Figure 2A [↗](#), S3A–B and Table S1). We then used a high-coverage transcriptomics dataset³¹ to ask how C_{sat} values relate to DFD gene expression in primary immune cells. Remarkably, transcript abundances significantly anticorrelated with C_{sat} in 17 of the 18 canonical immune cell populations (Figure 2B [↗](#) and S3C). Such a relationship is highly unusual for soluble proteins, whose expression instead tends to strongly correlate with C_{sat} .³² These analyses collectively indicate that discontinuous DFDs are likely to be functionally supersaturated in their endogenous physiological contexts.

Given that nucleation barriers are restricted to only a subset of DFDs, we asked if the DFDs with nucleation barriers tend to have different signaling roles than those without. From our prior demonstrations of prion-like activity by ASC, MAVS, and BCL10,^{12,33} we suspected an enrichment among adaptors. However, DFD proteins have not been systematically evaluated with respect to adaptor function. We therefore focused on the defining property of “adaptors” physically connecting other proteins by quantifying their centrality in the DFD protein-protein interaction

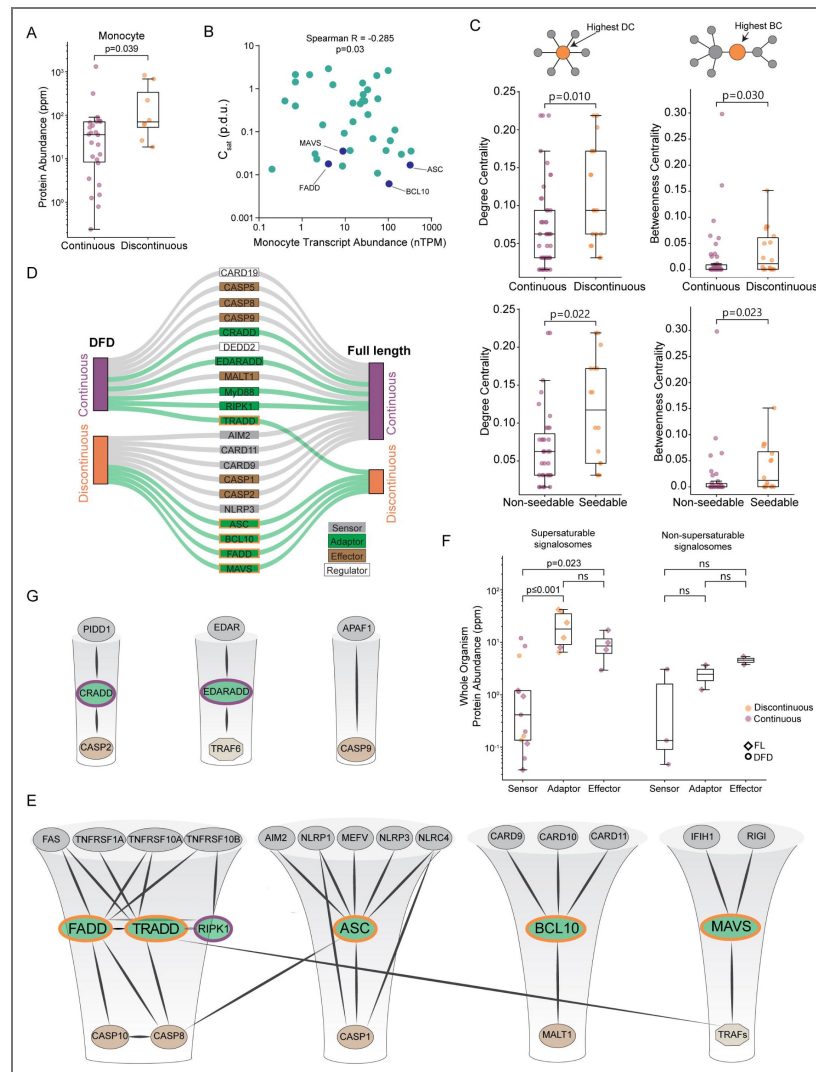


Figure 2. Nucleation barriers are a characteristic feature of inflammatory signalosome adaptors.

A. Boxplot of DFD-containing protein abundances in monocytes, showing that discontinuous DFDs have higher endogenous expression levels. Mann-Whitney U = 53, $n_{\text{continuous}} = 26$, $n_{\text{discontinuous}} = 8$ ($p = 0.039$). Protein abundance values are from PAXdb.²⁸ B. Scatter plot of DFD gene expression in monocytes (normalized transcripts per million) and C_{sat} values. Spearman R = -0.285 ($p = 0.03$). Adaptor DFDs are labeled. Dataset obtained from the Human Protein Atlas. C. Top: box plots of degree centrality (left) and betweenness centrality (right) of continuous and discontinuous DFDs in the endogenous network of physically interacting DFD proteins, showing that the latter are more centrally positioned. Degree centrality Mann-Whitney U = 242.0 ($p = 0.010$); betweenness centrality Mann-Whitney U = 274.0 ($p = 0.030$); $n_{\text{continuous}} = 46$, $n_{\text{discontinuous}} = 18$. Bottom: box plots of centrality measures of nonseedable and seedable DFDs, showing that the latter are more centrally positioned. Degree centrality Mann-Whitney U = 167.5 ($p = 0.022$); betweenness centrality Mann-Whitney U = 172.5 ($p = 0.023$); $n_{\text{non-seedable}} = 35$, $n_{\text{seedable}} = 16$. D. Visualization of how the DAMFRET profiles of isolated DFD domains (left) change in their full-length contexts (right), showing that only adaptor proteins (green connections) tend to retain discontinuous transitions in their full-length context. E. Subnetworks of prominent signalosome adaptor proteins that were found to be supersaturable. Edges connect nodes with experimentally determined physical interactions with confidence > 0.9 in STRING. All proteins shown have DFDs except TRAFs. Each adaptor's node size is proportional to its supersaturability score. F. Comparison of protein abundances at the whole body level for the signalosome components in Figure 2E (left) and Figure 2G (right), showing that adaptors are more highly expressed for the former. Protein abundance values are from PAXdb.²⁸ P-values are from Mann-Whitney test. For supersaturable signalosomes: $n_{\text{sensor}} = 13$, $n_{\text{adaptor}} = 6$, $n_{\text{effector}} = 4$; sensors and adaptors, U = 4.0 ($p < 0.001$); sensors and effectors, U = 6.0 ($p = 0.023$); adaptors and effectors, U = 18.0 ($p = 0.257$). For non-supersaturable signalosomes: $n_{\text{sensor}} = 3$, $n_{\text{adaptor}} = 2$, $n_{\text{effector}} = 2$; sensors and adaptors, U = 1.0 ($p = 0.400$); sensors and effectors, U = 0.0 ($p = 0.200$); adaptors and effectors, U = 0.0 ($p = 0.333$). G. Subnetworks of signalosomes lacking supersaturable DFDs. Edges connect nodes with experimentally determined physical interactions with confidence > 0.9 in STRING. All proteins shown have DFDs except TRAF6. See also Figures S3, S4 and S5, and Tables S1 and S2.

network. Proteins with high “degree centrality” are hub-like, with many direct interactions. Proteins with high “betweenness centrality” lie on the shortest paths between other nodes and therefore act as bottlenecks in the network. Assessing both measures (see methods), we observed that discontinuous and/or seedable DFD proteins have significantly greater degree and betweenness centralities (Figure 2C and S3D-E). As a point of comparison, we performed the same analysis for the domain family most closely resembling DFDs – Sterile Alpha Motif (SAM). SAM domains have similar size, structure, number (76 in humans), and function as DFDs, often serving as polymeric scaffolds of signaling complexes.^{34–36} Unlike DFDs, however, SAM domains cannot supersaturate.²⁴ Consistent with that, we found that they also have much lower betweenness centrality in their physical interaction network as compared to discontinuous DFDs ($p = 0.0003$; Figure S3F).

For a DFD nucleation barrier to enable on-demand assembly, the DFD must remain supersaturated even as a full-length (FL) protein. This means that other parts of the protein must not trigger nucleation prematurely, as through homo-oligomerization. Nor should they disfavor the assembled state, which would raise C_{sat} . We therefore evaluated the phase behaviours of 21 diverse DFD-containing FL multidomain proteins. Most of the proteins behaved the same way as their respective DFDs (Figure 2D, Figure S4A, and Table S2). Six proteins suppressed the ability of their DFDs to supersaturate, resulting in continuous low or moderate DAMFRET, suggesting they form non-nucleated oligomers. To determine if the DFDs are in an active polymer configuration within these oligomers, we tested the ability of FL NLRP3 to nucleate its cognate adaptor, ASC. It failed to do so (Figure S4B), revealing that the oligomers are “autoinhibited” as has been previously demonstrated for multiple DFD-containing receptors and effectors.^{9,37–43} This implies that the DFD within FL NLRP3 (and presumably other DFDs that lost supersaturability in their FL context) cannot self-assemble; instead, its joining the polymer structure is driven by the energy released by ligand (e.g. D/PAMP) binding. This stoichiometric requirement limits spurious activation that would otherwise kill or inflame cells unnecessarily.^{9,37–42} In contrast to these autoinhibited proteins, five of the FL proteins instead retained or enhanced their DFDs’ nucleation barriers: ASC, BCL10, FADD, MAVS, and TRADD (Figure 2D, Figure S4A, and Table S2). Testing a subset of FL proteins further, we found that only this latter group can be self-seeded (Figure S5A). These proteins collectively span all four DFD subfamilies, and all of them function as major adaptors in innate immune signalosomes (Figure 2E). ASC, BCL10, MAVS, and TRADD drive inflammation and/or inflammatory cell death downstream of D/PAMP sensing.^{44–50} FADD signals downstream of death ligands and certain D/PAMPs through various signalosomes that can be either anti- or proinflammatory.^{51–56} In contrast, signalosomes with primarily non-immunity functions – the apoptosome, the PIDDosome, and the ectodysplasin (EDA) receptor complex^{57–59} – lacked supersaturation (Figure 2G). Consistent with their functioning through supersaturation, only the innate immune set of adaptors are more abundantly expressed than their cognate receptors in the human body (Figure 2F). These data collectively identify the adaptors, specifically, of inflammatory signalosomes as energy reservoirs for signal amplification.

Nucleation barriers may facilitate signal amplification in human cells

To explore the functionality of supersaturation, we focused on two modes of programmed cell death signaling that differed in this regard: intrinsic apoptosis and pyroptosis. The former occurs downstream of persistent intracellular stresses.^{60–62} The latter is instead triggered by minute levels of D/PAMPs and therefore involves greater signal amplification. Our DAMFRET analyses revealed that CARDS of the apoptosome lack nucleation barriers whereas CARDS of the inflammasome, which drives pyroptosis, have nucleation barriers (Figure S6A).

To test if the absence of nucleation barriers limits the sensitivity of the apoptosome, we adapted an optogenetic approach^{12,63–65} to precisely control the initiation of intrinsic apoptosis. HEK293T cells lack the inflammasome constituents NLRC4, ASC, and CASP1, allowing us to repurpose their DFDs in this cell line. Accordingly, we transduced the cells with mScarlet-I fusions of either the non-supersaturable WT apoptosome effector, CASP9, or a chimeric version that harbored the

supersaturable DFD of CASP1 in place of its own (CASP9^{CASP1CARD}). We simultaneously transduced the cells with blue light-inducible seeds of the cognate upstream DFDs – opto-APAF1 or opto-NLRC4, respectively (Figure 3A). We then assessed signal propagation through the reconstituted pathways by measuring activation of CASP9's downstream target, CASP3/7, after one minute of blue light stimulation. Both cell lines activated a fluorescent CASP3/7 reporter to the same extent (Figure 3B), confirming that the different modifications do not sterically interfere with the proteins' activities.

To now assess signal amplification, we evaluated both the persistence of DFD assemblies and their ability to commit cells to apoptosis following an otherwise sublethal stimulus. We anticipated that cells expressing non-supersaturable CASP9 will 1) form transient clusters and consequently 2) survive a short blue light stimulus. In contrast, cells expressing supersaturable CASP9 will 1) form perdurant clusters that signal indefinitely, and consequently 2) die even after a short blue light stimulus.

To test prediction 1, we stimulated the cells with blue light for one minute and monitored subsequent protein localization. We found that both pairs of proteins formed puncta in essentially all cells within one minute (Figure 3C-D; Movie S1). The puncta with non-supersaturable CASP9 then dissolved over the course of the next 10 minutes. In contrast, the puncta with supersaturable CASP9 instead continued to grow at a constant rate for at least the next 20 minutes (Figure 3C-D; Movie S2), confirming that the protein's drive to polymerize exists even without the stimulus.

To test prediction 2, we measured cell death two hours after the initiation of blue light stimulation for either one minute or the entire two hours. Neither pair of proteins induced cell death in the absence of blue light (Figure 3E). Non-supersaturable CASP9 induced death in approximately 18% and 55% of cells following the short and long stimulation, respectively (Figure 3E), confirming the expected dose-dependence of signaling. In contrast, supersaturable CASP9 induced death in most cells even upon short stimulation, consistent with the expected amplification of signaling due to supersaturation-mediated DFD polymerization. We note that the increased stability of NLRC4^{CARD} and CASP1^{CARD} polymers relative to APAF1^{CARD} and CASP9^{CARD} oligomers could increase signal amplification irrespective of polymerization however, due to the slower dissolution of assemblies that formed even during stimulation, which would give CASP9^{CASP1CARD} more time to activate more molecules of CASP3/7. Consequently, further work will be required to distinguish the contributions of each mechanism to amplification.

Some innate immune adaptors are endogenously supersaturated

We next asked if pyroptosis and extrinsic apoptosis are indeed powered by adaptor supersaturation in vivo. We first induced pyroptosis in human THP-1 monocytes by treating them with poly(dA:dT), a ligand for the inflammasome receptor, AIM2. By 18 hours, approximately 70% of the cells were dead or dying. We then deleted *PYCARD* to determine if cell death depended on ASC, the adaptor for the inflammasome. Death was delayed, but not eliminated (Figure 4A). We then also deleted *FADD*, encoding an adaptor that has been shown to co-assemble with AIM2 in pathogen-triggered PANoptosis.⁶⁶ Death was reduced further still (Figure 4A and S6B), confirming that AIM2 activation can induce cell death through both ASC and FADD.

To eliminate the potentially confounding effects of orthogonal dsDNA sensors,⁶⁷ we next placed AIM2^{PYD} under blue light control and confirmed that the resulting fusion protein – “opto-AIM2” – grants direct control over seed formation (Figure 4B and 4C, top). Using this system, a transient blue light exposure induced AIM2 clustering within seconds and cell death within ten minutes (Figure 4C-D and Movie S3-S4). Next, we evaluated if cell death depended on ASC or FADD. We found that stimulating WT cells with a one-second blue laser pulse every 15 minutes killed essentially all of them within one hour (Figure 4E). In contrast, only half of cells lacking ASC (*PYCARD*-KO) died, and they did so with delayed kinetics consistent with FADD-driven signaling.^{52,68,69}

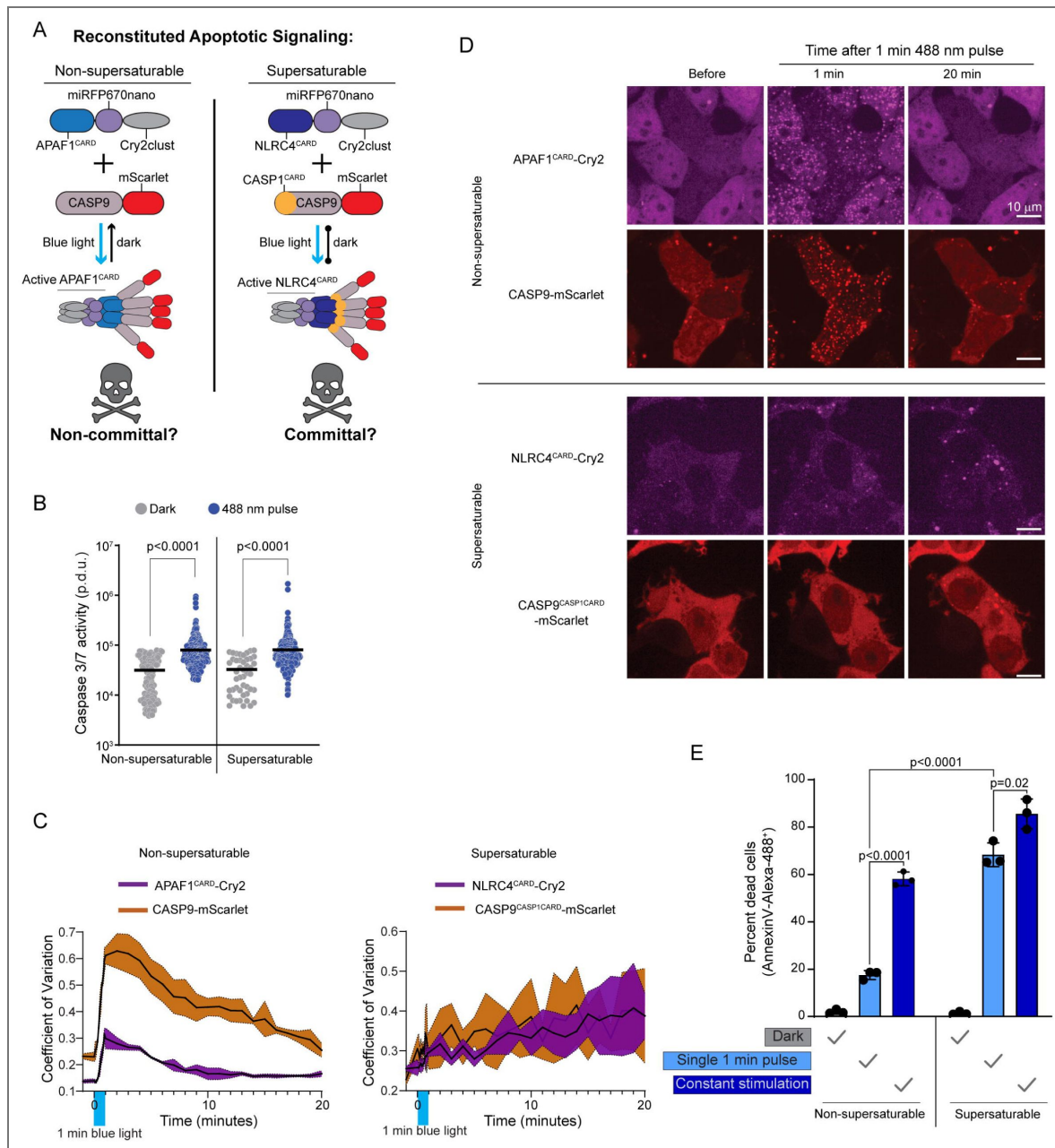


Figure 3. Nucleation barriers may facilitate signal amplification in human cells.

A. Schematic diagram of experiment in HEK293T cells to reconstitute the apoptosome with optogenetic control, in either a non-supersaturable or supersaturable format. The nonsupersaturable format comprises CASP9 activated by APAF1^{CARD} (as in the native apoptosome); the supersaturable format comprises chimeric CASP9 with CASP1^{CARD} replacing CASP9^{CARD} (CASP9^{CASP1^{CARD}}), activated by chimeric APAF1 with NLRC4^{CARD} in place of APAF1^{CARD}. Blue light triggered assembly in both cases, but subsequent continued assembly in the dark only occurred for the supersaturated format. B. Caspase 3/7 activity reporter fluorescence intensities in the absence of stimulation or after one minute of 488 nm stimulation for cell lines expressing the non-supersaturable or supersaturable pairs, showing that both pairs comparably activate caspase 3/7 while oligomerized. APAF1^{CARD}-Cry2 + CASP9-mScarlet-I, dark $n = 163$, pulse $n = 375$, Mann-Whitney $U = 11362$ ($p < 0.0001$). NLRC4^{CARD}-Cry2 + CASP9^{CASP1^{CARD}}, dark $n = 46$, pulse $n = 305$, Mann-Whitney $U = 4253$ ($p < 0.0001$). C. Coefficient of variation (CV) of fluorescence distribution in HEK293T cells expressing the indicated protein pairs after a single one minute 488 nm laser activation. Top, APAF1^{CARD}-Cry2 and CASP9-mScarlet-I display rapid cluster formation that dissociates by 20 min. Bottom, NLRC4^{CARD}-Cry2 and chimeric CASP9^{CASP1^{CARD}} cluster less rapidly but the clusters continue to grow indefinitely. D. Representative images from experiment in C. Clusters of APAF1^{CARD}-Cry2 and CASP9-mScarlet-I form then dissociate while NLRC4^{CARD}-Cry2 and CASP9^{CASP1^{CARD}} clusters only get larger. E. Quantification of cell death of the HEK293T chimeric cells (as in A) using Annexin V-Alexa 488 staining, either two hours after a single one minute pulse of 488 nm laser, or after two hours of "constant" stimulation whereby cells were subjected to a one second pulse every one minute. P-values derived from t-test. See also Figure S6, Table S3 and Movie S1, S2.

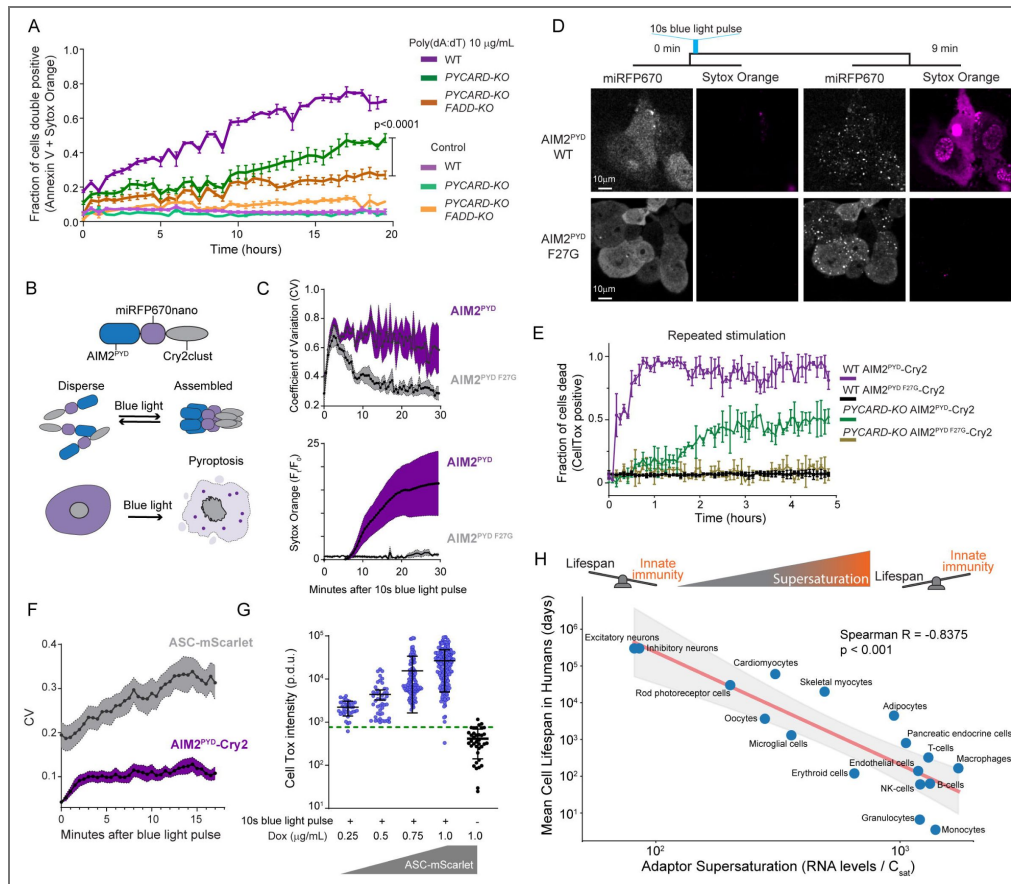


Figure 4. Innate immune adaptors are endogenously supersaturated.

A. Time course of apoptotic cell death of THP-1 cells following exposure to AIM2 ligand, poly(dA:dT). P-value obtained from ANOVA followed by pair comparison. B. Schematic diagram of the experiment to transiently optogenetically stimulate AIM2^{PYD} to monitor ASC^{PYD} assembly. This experiment was conducted in HEK293T cells because they do not undergo pyroptosis. C. Top, Time course of fluorescence intensity distribution in THP-1 cells following 10 seconds of optogenetic activation, showing that WT AIM2^{PYD} forms clusters (high CV) that persist and induces cell death, while the F27G solubilizing mutant²³ forms clusters that subsequently disperse. Bottom, normalized Sytox Orange fluorescence intensity for the experiment in the top panel. D. Representative confocal microscopy images from a timelapse of THP-1 monocytes showing that transient optogenetic stimulation of WT but not F27G mutant of AIM2^{PYD} causes it to form puncta that coincide with cell death. Sytox Orange was used for this experiment because it can be excited without activating Cry2. E. Time course of cell death of THP-1 cells when subjected to a blue light pulse every 5 minutes ("repeated"), showing rapid cell death (violet trace) only when AIM2^{PYD} is WT and when ASC is present. The absence of ASC results in slower death (green trace), consistent with apoptosis. The F27G mutation of AIM2^{PYD} blocks cell death irrespective of ASC (black and golden traces). F. Coefficient of variation (CV) of fluorescence distribution of AIM2^{PYD}-Cry2 and ASC-mScarlet-I in THP-1 PYCARD-KO cells following a 10 s blue light pulse. This shows that AIM2^{PYD} and ASC-mScarlet-I (with slightly delayed kinetics) rapidly form clusters that persist well after stimulus removal. ASC-mScarlet-I was induced to only ~20% of the ASC expression in WT cells using 1.0 µg/mL doxycycline (dox). G. Quantification of CellTox staining in individual ASC-mScarlet-I THP-1 PYCARD-KO cells 30 minutes after a 10 second blue laser pulse, at different levels of dox-induced ASC-mScarlet-I expression. Green dotted line indicates 95% confidence interval (CI) for background fluorescence intensity, above which cells were considered CellTox-positive. Error bars denote standard deviation. Control, n = 37. 0.25 µg/mL dox, n = 36. 0.5 µg/mL dox, n = 47. 0.75 µg/mL dox, n = 113. 1 µg/mL dox, n = 180. H. Top: The metastability of supersaturation implies that cells will occasionally inflame and/or die from stochastic (without D/PAMPs) DFD nucleation, which creates a tradeoff between innate immunity and life expectancy. Bottom: Scatter plot showing the relationship between geometric mean of adaptor supersaturation including ASC, FADD, BCL10, TRADD, MAVS (as approximated by the ratio of transcription levels and C_{sat} values) and mean lifespan for each cell type in the human body for which data is available.⁷³ Cell types with greater DFD supersaturation have shorter mean lifespans. The red line represents the best-fit power-law regression, obtained by performing linear regression in log-log space. The shaded region represents the 95% confidence interval for the trend line. Spearman R = -0.8375 (two-tailed p = 0.000027). See also Figure S6 and Table S3.

Finally, we directly assessed inflammasome nucleation by AIM2^{PYD} seeds. To do so we reconstituted the *PYCARD*-KO with mScarlet-I-tagged ASC and titrated its expression to well below endogenous levels so as to circumvent any potential for the fusion tag to enhance the protein's oligomerization (although this would be unexpected,⁷⁰ Figure S6C-E). We then tracked AIM2^{PYD} and ASC localization following a 10-second laser pulse. Both proteins began to cluster almost immediately. AIM2^{PYD} cluster growth nearly stopped by three minutes, consistent with excited Cry2 relaxation on a similar time scale.^{63,71} In contrast, ASC clusters continued rapid growth for more than ten minutes (Figure 4F [↗](#)), demonstrating that their drive to do so is independent of the triggering stimulus. The plasma membrane concomitantly permeabilized as a result of gasdermin D (GSDMD) activation,⁷² and to an extent that increased with the level of ASC expression (Figure 4G [↗](#)). These data collectively confirm that endogenous ASC is highly supersaturated prior to stimulation, and that the extent of supersaturation determines the extent of signal amplification.

Given that inflammatory signalosomes frequently initiate programmed cell death, and that a kinetic barrier governs their activity, the susceptibility of cells to aberrant cell death through spontaneous nucleating fluctuations is expected to increase with the level of adaptor supersaturation in vivo. This would constrain a cell's life expectancy if such nucleation occurs with appreciable frequency. To explore this possibility, we asked how adaptor supersaturation (as approximated by the ratio of mRNA levels to C_{sat} values) relates to the turnover rates of each cell type in the human body.⁷³ We found that short-lived cells such as monocytes indeed have greater DFD supersaturation than long-lived cells such as neurons (Figure 4H [↗](#)). Among individual DFD proteins, cell turnover correlated especially strongly with the expression level of ASC (Figure S6F-G), suggesting that life expectancy may be limited by the thermodynamic drive for inflammatory signal amplification.

The nucleating interactome is highly specific

DFDs share the same fold and are co-expressed in many of the same cells. While spontaneous nucleation of supersaturable DFDs poses an inherent risk of inadvertently activating end fate cell signalling, this possibility is multiplied by the risk of cross-activation between DFDs in different pathways. To what extent do they nucleate each other? To determine specificity and systematically map the nucleating interactome of DFDs, we mated our library of seed-expressing yeast strains with a library of strains expressing each mEos3-fused DFD (Figure S7A) to create over 10,000 arrayed diploid yeast strains that we then screened by DAMFRET (Figure S7B and C).

In total, we identified 171 nucleating interactions, representing just ~1.6% of the total library (Figure S7D-F and Table S4). The interactions were largely constrained not just to members of the same subfamily, but to members of the same signaling subnetwork even within subfamilies (Figure 5A-B [↗](#)). For example, CARDs of the CBM signalosome nucleated each other but not CARDs of the inflammasome, and vice versa. As an exception, PYDs of the inflammasome and DEDs of the Death-Inducing Signaling Complex (DISC) nucleated each other (Figure 5C [↗](#)). These interactions are consistent with the residual cell death we observed in the *PYCARD FADD* double knockout, as well as previously observed crosstalk between these pathways and the close phylogenetic relationship between PYDs and DEDs.^{69,74–76}

On the whole, the observed network of nucleating interactions reveal that DFDs from different pathways generally function independently of each other, allowing adaptors to serve as orthogonal energy reservoirs for their respective signalosomes (as schematized in Figure 2E [↗](#)).

DFD nucleation barriers are deeply conserved

Animals acquired DFDs horizontally from bacteria,^{5,77,78} giving rise to an ancestral DISC (conserved across metazoa) and later inflammasome (conserved across vertebrates; Figure 6A [↗](#)).^{79,80} To investigate conservation of nucleation barriers, we characterized by DAMFRET the phase behaviours of DFD-containing constituents of a basal inflammasome – NLRP3, ASC, and CASP1 from zebrafish (*Danio rerio*); and a basal DISC – A0A1X7U321, FADD, and CASP8 from the model sponge, *Amphimedon queenslandica*. As for their human counterparts, the fish and sponge

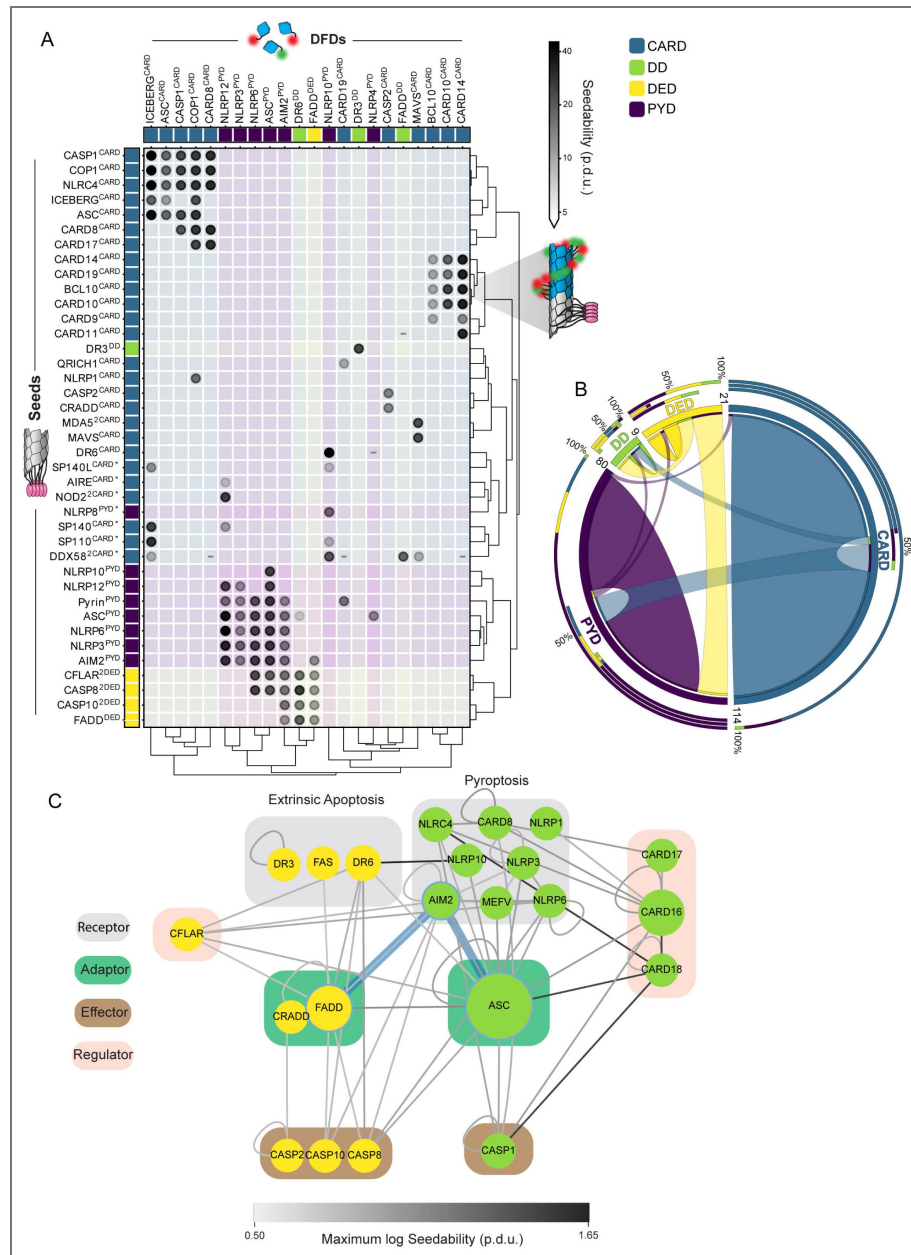


Figure 5. The nucleating interactome is highly specific.

A. Matrix of all nucleating interactions (gray-shaded circles) detected in a comprehensive DAMFRET screen of > 10,000 DFD pairs. Each DFD-mEos3 (columns) was separately expressed with each DFD- μ NS seed (rows). Darker shading of the circle denotes increased seedability. Interactions among members of the same signaling pathway (in legend) appear in color shaded squares. Asterisk denotes seeds that were screened in a separate experiment from the rest. The matrix was clustered on seedability values, on a log scale, using the SciPy.cluster.hierarchy v1.11.1 linkage and dendrogram Python packages, using the Ward variance minimization algorithm to calculate distances. Procedure defined units (p.d.u.). B. Circos plot of the nucleating interactions summarized by DFD subfamily. Each subfamily is represented with a segment proportional to the number of DFDs with a nucleating interaction, as indicated by ribbons within and between segments. Inner stacked bars around the perimeter show the numbers of DFDs in each subfamily seeded by the subfamily in that segment. Middle stacked bars around the perimeter show the numbers of DFDs that seed the subfamily in that segment. Outer stacked bars around the perimeter show total nucleating interactions involving the subfamily in that segment. C. Nucleating interactions involving DFDs in extrinsic apoptosis and pyroptosis, with blue edges highlighting the direct nucleating effect of AIM2 on FADD and ASC that is explored in Figure 4. The network was created in Cytoscape with node size corresponding to betweenness centrality and grouped by reported function. Interactions between FL proteins (Table S2) were included. Edge darkness indicates the seedability score of the corresponding interaction. See also Figure S7 and Table S4.

adaptors -- but not the receptors and effectors -- were supersaturable (Figure 6B [↗](#) and S8A-B). These results suggest that the function of DFDs as energy reservoirs preceded the radiation of animals.

Discussion

Our systematic investigation revealed that metastable supersaturation of a select subset of adaptor proteins provides an energetic basis for DFD-mediated signal amplification.

Seminal studies on the structures of DFDs have led to the common view that DFDs generally function as homotypic interaction modules.^{9,81} We were therefore surprised to find that approximately half (51) of human DFDs failed to detectably self-assemble even when expressed well-beyond physiological concentrations. This suggests that a large fraction of DFDs function through hetero- rather than homo-oligomerization. From known examples we postulate these will involve 1) receptors templating adaptor nucleation, 2) effectors binding to adaptor polymers, and 3) regulators repressing signaling by competing with 1 and 2.^{82–85} Importantly, and despite a general appreciation that DFD polymerization functions to amplify signaling, most of the DFDs that self-assembled (36 of 57) lacked sufficient nucleation barriers to supersaturate at the cellular level. This precludes them from amplifying signals without additional energy expenditure by the cell. The only DFDs to supersaturate in full-length protein contexts were adaptors of inflammatory signalosomes. The sparsity of this property belied its functional importance, however, as indicated by higher centrality measures. Generally, the more central proteins of networks tend to be essential,⁸⁶ more abundant,⁸⁷ slower evolving,⁸⁸ and more frequently targeted by pathogens than noncentral proteins.^{87,89} Examining the topology of DFD networks revealed a core set of signal amplifying adaptors each interacting with a sequence-specified set of receptors, effectors, and regulators that categorically lack nucleation barriers. Because nucleation involves only a tiny fraction of the supersaturated adaptors, they are poised to amplify minute signals from their cognate receptors and transduce them to their cognate effectors, precisely as required for innate immunity. Furthermore, because this attribute is conserved across homologous animal adaptors, energy storage for on-demand signaling may be an ancestral function of DFDs.

If supersaturation is indeed important for innate immune signaling, why are so few DFDs supersaturable? The answer may lie in the fact that each supersaturated node in a death-inducing pathway imposes a risk of unintentional death. We speculate that evolution therefore minimizes the number of supersaturated DFDs by restricting them to only the central nodes of the network. In this way, a restricted number of supersaturable DFDs can be continuously “repurposed” with receptor and effector proteins specific to each D/PAMP and mode of cell death.

Two prominent inflammatory signaling adaptors -- MyD88 and RIPK1 -- defied the broader trend in their inability to supersaturate. Intriguingly however, each functions in series with other adaptors that *do* supersaturate. MyD88 typically functions downstream of TIRAP,⁹⁰ whose TIR domain drives nucleation-limited polymerization in vitro and in cells.⁹¹ Similarly, RIPK1 acts in conjunction with TRADD leading to either NF- κ B activation, apoptosis (via FADD), or necroptosis (via RIPK3). TRADD and FADD were shown here to supersaturate, while RIPK3 contains an amyloid-forming motif (RHIM) whose self-assembly is required for necroptosis.⁹² Hence, all of the partner adaptors of MyD88 and RIPK1 are either demonstrably or plausibly supersaturable, thereby obviating such functionality in MyD88 and RIPK1. Consequently, these exceptions in fact support the generalization that each inflammatory pathway has approximately one supersaturable node. Alternatively, MyD88 may be a false negative in our experiments, perhaps due to an absence of a required factor in yeast cells or the presence of a nucleating factor that does not exist in human cells. Unlike other adaptors which are effectively monomeric prior to activation, MyD88 forms discrete oligomers that activate by clustering into larger assemblies.^{93–95} DAmFRET may be unable to resolve the transition between these multimeric forms. MyD88 was previously found to exhibit a nucleation barrier in vitro,⁹⁶ and to form switch-like and persistent assemblies at the subcellular level in vivo.^{95,97}

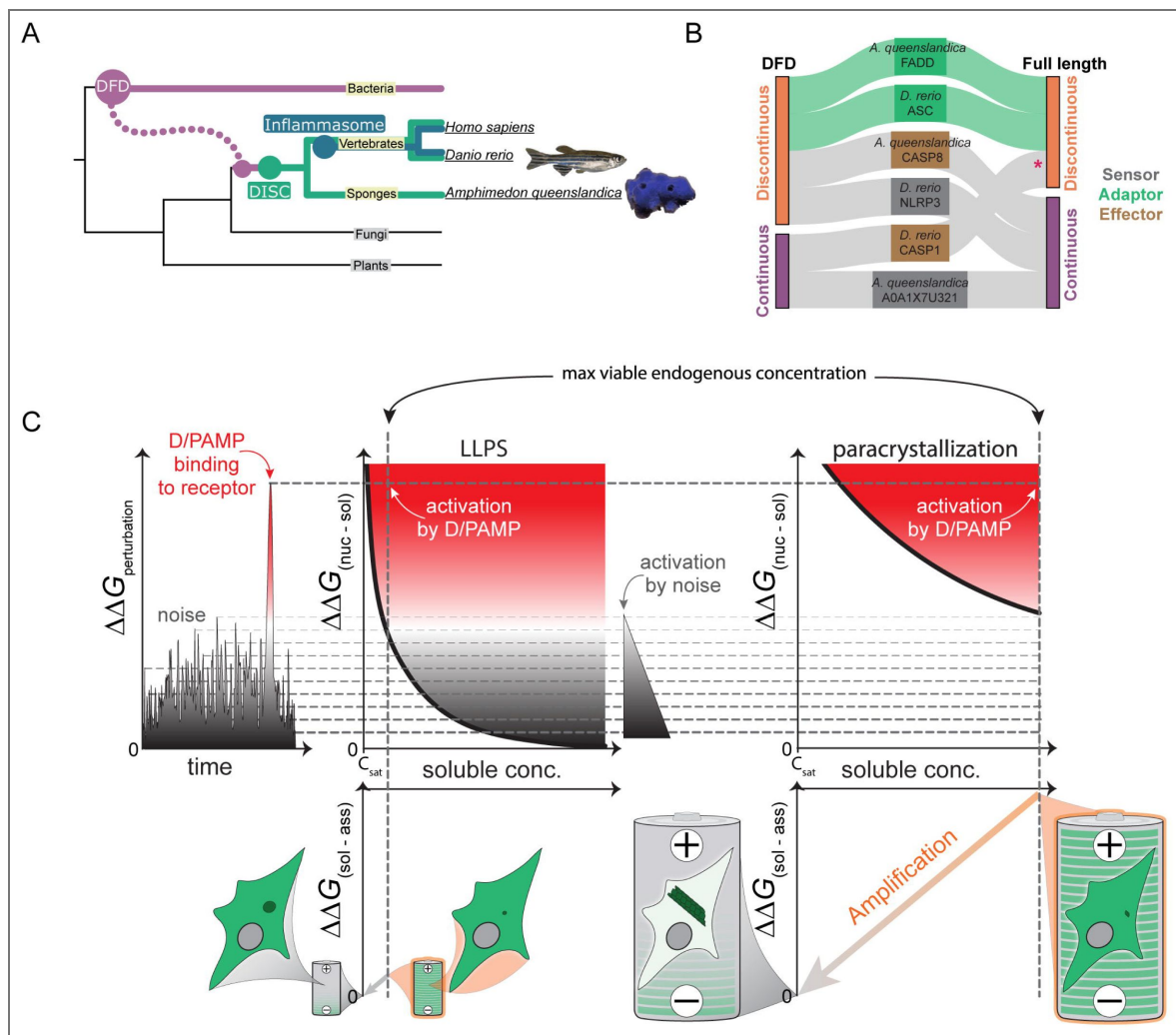


Figure 6. DFD nucleation barriers are deeply conserved.

A. Phylogenetic tree illustrating evolutionary relationships between DFD signaling pathways from bacteria to humans. B. DAMFRET classifications for DFD-only and FL components of the DISC from the model sponge, *Amphimedon queenslandica*, and of the inflammasome from the model fish *Danio rerio*, showing that adaptors are specifically supersaturable. **D. rerio* CASP1^{FL} exhibits a high C_{sat} in the mid-micromolar range, which greatly exceeds the nanomolar concentration expected for endogenous procaspase^{1,118}, making it unlikely to supersaturate at endogenous concentrations. C. Physical logic of DFD function. **Left:** Cells experience thermodynamic perturbations either from stochastic fluctuations (noise) or D/PAMP binding to innate immune receptors. These perturbations can nucleate supersaturated signaling proteins (dashed horizontal lines) with a probability that depends on the type of phase transition and specifically, whether it is accompanied by structural ordering. **Middle:** For phase separation in the absence of structural ordering (LLPS), the nucleation barrier ($\Delta\Delta G_{(nucleus - solute)}$) declines sharply with concentration beyond C_{sat} ^{103,119}, which increases its susceptibility to noise. This limits the level of supersaturation that can be maintained by a cell (vertical dashed line), and therefore, the extent to which assembly ($\Delta\Delta G_{(solute - assembly)}$) can power signal amplification (tiny battery schematic). **Right:** For phase separation with structural ordering (paracrystallization as in adaptor DFD assemblies), the dependence of nucleation on concomitant intramolecular fluctuations buffers the barrier against concentration (as indicated by a shallower curve relative to LLPS), which allows cells to maintain much higher levels of supersaturation^{16,21}. Following nucleation, the assemblies grow and deplete soluble protein until it is no longer supersaturated, driving amplification (diagonal orange arrow) through proximity-dependent effector activation. The intrinsic nucleation barriers encoded by solution phase DFD ensembles therefore allow them to function as phase change batteries (giant battery schematic) to power innate immune signal amplification. See also Figure S8 and Table S2.

We liken supersaturated DFD adaptors to ‘phase change materials’ in industrial thermal batteries, accumulating potential energy for subsequent deployment upon nucleation.^{98,99} This differs fundamentally from signaling cascades that rely on chemical fuels like ATP. While ATP is a common good shared across cellular processes, DFD batteries are autonomous power sources, exclusive to each signaling pathway and therefore insulated from other cellular processes. This could allow for signal transduction independently of variable or compromised cell metabolism. Just as a battery cannot recharge itself, the DFD assemblies are effectively irreversible, committing the cell to terminal responses like death. This battery function rationalizes DFD prevalence over more common assembly modes like liquid-liquid phase separation (LLPS; Figure 6C [↗](#)). While LLPS offers sensitivity that can enhance D/PAMP detection,^{100,101} the sharp drop in its nucleation barrier with increasing supersaturation^{102–104} makes it unsuitable for long-term, high-energy storage needed for explosive signal amplification through rapid assembly.

Signalosome effectors activate through simple induced-proximity mechanisms that lack specific structural requirements of the DFDs themselves,^{12,64,105,106} which suggests that the structures of DFD polymers are determined by a function of the solution phase *prior to activation*. Our finding that the function is energy storage rationalizes the strikingly regular structure of DFD polymers. The evolution of this function would have been driven by selection *against* premature assembly, favoring paracrystalline ordering whose infrequent nucleation allows for energy storage over a cell’s lifetime. This differs from selection *for* a specific functional polymer structure, as seen for example in microtubules. This principle may also explain the functional substitution of DFDs with alternative paracrystalline modules like TIR domains and amyloids in some innate immune signaling pathways,⁸ as well as the dearth of non-nucleated polymers, such as those of SAM domains, in innate immune signaling despite their prevalence in other signaling pathways.³⁴

Despite functional nucleation barriers potentially driving the evolution of DFD polymerization, we emphasize that nucleation barriers are directly determined by the soluble ensembles that precede polymers. Our finding that only a subset of polymerizing DFDs exhibit nucleation barriers underscores this fact. These non-nucleation-limited polymers presumably have larger energetic differences between their orthogonal interfaces than those of nucleation-limited polymers, which would be expected to reduce the cooperativity of nascent polymer formation.¹⁰⁷ Fully elucidating the DFD structure-function relationship will require future exploration of the conformational preferences of monomers and early stage oligomers.

The ability of a protein fold to crystallize has few structural constraints, allowing relatively small changes in sequence to produce orthogonal signaling modules. The resulting evolvability may be essential in the never-ending arms race against pathogens. High specificity also insulates pathways from each other and from cellular processes and metabolic fluctuations that could aberrantly activate them to lethal consequence.^{108–110} The precise determinants of specificity will differ for each interaction, but prior work has uncovered principles that will likely prove general, such as the shape and electrostatic complementarity of interfaces and cross-compatibility of the polymers’ helical architectures.^{23,83,111–113}

Our findings imply that cells perpetually await death. The theoretical cumulative certainty of stochastic nucleation over time appears to be reflected in the observed relationship of DFD supersaturation to mortality rates across human cell types. We speculate that this underpins a fundamental tradeoff between innate immunity and life expectancy, potentially contributing to age-related inflammation and stem cell exhaustion.¹¹⁴

Limitations of the study

Multiple DFDs were found to populate stable ordered polymers in all cells. Although we did not observe a nucleation barrier for these, it is possible that they are supersaturable *in vivo* at much lower than the micromolar concentrations surveyed by DAmFRET. Similarly, a small number of DFDs that populated only a low AmFRET state did not express to high concentrations in yeast, and it is possible that they can self-assemble under cellular contexts that allow them to reach higher concentrations.

Our study necessarily simplifies and abstracts human innate immune signaling. We considered the cytoplasm of living yeast cells to be a suitable proxy for the physiological context of DFDs *as a whole*, while acknowledging that the endogenous contexts of DFDs differ from that in our experiments. For example, several of the DFDs normally localize to membranes or the nucleus in human cells, and this could impact their phase behavior. Nevertheless, we find no evidence that such localizations would reduce supersaturation, as none of the DFDs from nuclear proteins exhibited nucleation barriers in our experiments. Likewise, in the case of MAVS, which normally localizes to the mitochondrial membrane, we found that the FL protein (containing its mitochondrial localization signal) was more supersaturable than its DFD alone, suggesting that localizing the protein to the mitochondrial surface increases the nucleation barrier despite restricting its diffusional entropy and increasing its local concentration. Our work also does not explore the many ways that DFD assembly can be regulated transcriptionally and posttranslationally. For example, NLRP3 is upregulated and becomes phosphorylated in response to LPS “priming” of the inflammasome,¹¹⁵ and polyubiquitination of RIG-I promotes its activation of MAVS.¹¹⁶ Such modifications imply that some signal amplification can occur upstream of supersaturated adaptors. The level of adaptor supersaturation differs between cell types (as we have shown) and is also likely to be dynamically regulated.

Methods

Reagents and antibodies

Hygromycin B (Invivogen, ant-hg-1), Penicillin-Streptomycin (ThermoFisher, 1514014gp), PMA (BioVision, 1544-5), Puromycin (Invivogen, ant-pr-1), Sytox Orange (ThermoFisher, S11368), CellTox (Promega, G8741), Annexin V Alexa568 (1:200, ThermoFisher, A13202), Annexin V Alexa488 (1:200, ThermoFisher, A13201), Incucyte® Caspase-3/7 Dye (Sartorius, 4440). Antibodies, anti-ASC (Santa Cruz Biotechnology, sc-514414), anti-FADD (Sigma, 05486), anti-Actin (Santa Cruz Biotechnology, sc-8432) were obtained from the indicated vendors.

Structural analyses

Twelve human proteins contain two DFDs, typically one closely following the other. To determine if the DFDs in such pairs should be evaluated independently or together, we used the predicted aligned error (PAE) matrix generated by AlphaFold3.¹²⁰ PAE is the expected positional error at residue x if the predicted and actual structures are aligned on residue y. Seven of the DFD pairs exhibited very low interdomain PAE scores comparable to those of the component DFDs (Figure S1B, bottom), suggesting a conserved fixed geometric relationship between the domains. We therefore considered these tandem DFDs as single members of their respective subfamilies. Similarly, we excluded the annotated⁶ PYD-like domain of CENP-N because the PAE matrix and experimental structures show that it is in fact part of a larger non-DFD.

Plasmid construction

Yeast expression plasmids were made as previously described.¹⁶ Briefly, we used a high copy episomal vector, V08, which contains inverted BsaI sites to support Golden Gate cloning, followed by a rigid helical linker 4x(EAAAR) and mEos3.1. This vector drives the expression of proteins from a *GAL1* promoter and contains the auxotrophic marker *URA3*. The vector V12 is identical to V08 except that mEos3.1 and linker precedes rather than follows the BsaI sites, for expressing proteins with an N-terminal fusion. Inserts were ordered as yeast codon-optimized GeneArt Strings (Thermo Fisher) flanked by Type IIs restriction sites for ligation between BsaI sites in V08 and V12. Fusions were made opposite the native N- or C-terminus of each DFD to minimize non-native steric effects. All other inserts were cloned into respective vectors via Gibson assembly between the promoter and respective tag. All plasmids were verified by Sanger sequencing. All expression plasmids are listed in Table S1.

Lentivirus vectors were as previously described.¹² Briefly, optogenetic constructs were cloned into pLV-EF1a-IRES-Hygro (Addgene #85134) which encodes a hygromycin B resistance cassette. To create lentiviral vectors expressing the the optogenetic constructs fused with miRFP670nano¹²¹ and Cry2, the corresponding sequences of AIM2^{PYD}, APAF1^{CARD}, NLR4^{CARD} were inserted via Gibson assembly into pLV-EF1a-IRES-Hygro. Finally, The doxycycline-controlled lentiviral vectors were cloned via Gibson assembly with the respective coding sequences from PYCARD, CASP9, CASP1, and mScarlet-I into pCW57.1 (Addgene #41393). All lentivirus vectors are listed in Table S3.

Yeast strain construction

Unseeded DAMFRET experiments were conducted using strain rhy1713.¹⁶ To create strains expressing DFD seeds, we first transformed AseI digests of each DFD plasmid along with a plasmid expressing Cas9 and a guideRNA targeting the *URA3* markers into rhy2153. This strain contains a genomic landing pad consisting of natMX followed by the *tetO7* promoter and counterselectable *URA3* ORFs derived from *C. albicans* and *K. lactis*, and stop- μ NS-mCardinal as described.¹² Successful integration of the insert replaces the *URA3* marker with the gene of interest and fuses to the protein's C-terminus μ NS-mCardinal, under the control of a doxycycline-repressible promoter. Transformants were selected for resistance to 5-FOA and validated for successful seed integration by detection of mCardinal expression using flow cytometry. The arrayed library of resulting strains was then mated to each of the rhy1713 strains expressing separate DFD-mEos3.1 fusions, by pinning each pair of strains together onto agar omnitrays containing SD-URA+NAT+dox media. The resulting colonies were then pinned into liquid SD-URA+NAT+dox for continued diploid selection and creation of glycerol stocks. The entire nucleating interaction screening consisted of 384 96-well plates.

DAMFRET assay preparation and data collection

We performed DAMFRET as previously described.¹⁶ Briefly, single transformant yeast colonies were inoculated in 200 μ L of SD-URA in a 96-well microplate well and incubated in a Heidolph Titramax platform shaker at 30°C, 1350 RPM overnight. Cells were washed with sterile water, resuspended in galactose-containing media, and allowed to continue incubating for approximately 20 h. Microplates were then illuminated for 25 min with 320–500 nm violet light to photoconvert a fraction of mEos3 molecules from a green (516 nm) form to a red form (581 nm). At this point, cells were either used to collect microscopy data or continue the DAMFRET protocol.

For the nucleating interaction screen, glycerol stock plates were pinned into liquid SD-URA without dox and incubated for 16 h at 30°C with 1350 RPM shaking overnight. We then resuspended cells in fresh SD-URA media and continued incubation for an additional 20 h. After this, we resuspended cells in SGal-URA and continued incubation for 20 h to induce protein expression. Finally, we resuspended cells in fresh SGal-URA for 4 h prior to DAMFRET data collection. The library was then consolidated into 96 384-well plates.

DAMFRET data were collected on a ZE5 cell analyzer cytometer. Autofluorescence was detected with 405 nm excitation and 460/22 nm emission; side scatter (SSC) and forward scatter (FSC) were detected with 488 nm excitation and 488/10 nm emission. Donor and FRET fluorescence were detected with 488 nm excitation and 425/35 nm or 593/52 nm emission, respectively. Acceptor fluorescence was detected with 561 nm excitation and 589/15 nm emission. For each well, we collected a volume of 13 μ L, resulting in approximately 500,000 events per sample. Data compensation was done in the built-in tool for compensation (Everest software V1.1) on single-color controls: non-photoconverted mEos3.1 and dsRed2 (as a proxy for the red form of mEos3). For nucleating interactions, we included an additional channel for mCardinal intensity with 561 nm excitation and 670/30 nm emission.

DAMFRET data analysis

Data were processed on FCS Express Plus 6.04.0015 software (De Novo). Events were gated for single unbudded cells by FSC vs. SSC, followed by gating of live cells with low autofluorescence and positive donor and acceptor fluorescence. With the exception of TNFRSF10A^{DD} (TRAIL-R1)

(rhx2933) which failed to express with either its C- or N-terminus tagged, all expression plasmids were processed. Plots represent the distribution of AmFRET (FRET intensity/acceptor intensity) vs. acceptor intensity (protein expression).

We then analysed the data as previously described.²⁰ Briefly, FCS files were gated using an automated R-script running in flowCore. Before gating, the forward scatter (FS00.A, FS00.W, FS00.H), side scatter (SS02.A), donor fluorescence (FL03.A), and autofluorescence (FL17.A) channels were transformed using a logicle transform in R. Single cells were gated using FS00.A vs SS02.A and FS00.H vs FS00.W. These were gated for expressing cells using FL03.A vs FL17.A. Cells falling within these gates were then exported as FCS3.0 files for further analysis.

DAmFRET histograms were divided into 64 logarithmically spaced bins across a predetermined range large enough to accommodate all data sets. The upper gate values were determined for each bin as the 99th percentile of the DAmFRET distribution in that bin. We used the expression of mEos3.1 alone to delineate the region of a DAmFRET plot that corresponds to no assembly. For all samples, cells falling above this region are considered to contain protein assemblies (FRET-positive). The fraction of cells in the assembled population was plotted as a ratio to the total cells in the bin for all 64 bins. The gross fraction of such cells expressing a given protein is reported as fgate.

Determination of continuity

We initially attempted to classify each DAmFRET dataset as one-state or two-state, and discontinuous or continuous for the latter, using an algorithm previously developed for this purpose.²⁶ However, the algorithm invariably misclassified discontinuous datasets as “continuous” when the AmFRET level of the high FRET state changed with concentration, i.e. exhibited positive or negative slope as for FAS^{DD} and CASP2^{CARD}, respectively. Due to this limitation, we adopted a different method.

To determine the continuity of an adequately expressed plasmid, we analysed the distribution of AmFRET values about the transition region. To do so, we fit a spline to the median AmFRET values across binned concentrations that met a density cutoff defined by a minimum of 20 cells, a minimum cell density of 500, and no more than 25% of cells being defined as an outlier. The bin density is measured by the number of cells divided by the interquartile range (IQR) of AmFRET values within that bin. The number of bins was determined using Scott’s rule for number of bins in a histogram. The median AmFRET values were calculated for each bin, and then denoised using the Python SciPy.signal.sosfiltfilt, resulting in the spline fit. This was bootstrapped 100 times with the average of all bootstraps reported as the final spline fit. The resulting spline was used to determine the transition region. The transition point is defined as the concentration with the greatest change in AmFRET. Therefore, this value was calculated as the maximum of the first derivative of the fitted spline, using the numpy.diff package. The transition range is defined as the region between the maximum positive and negative rate of change, indicating where the transition between FRET states is starting and ending, respectively. The transition starting point is calculated as the maximum of the second derivative that lies before the transition point, and the transition endpoint is calculated as the minimum of the second derivative that lies after the transition point. This is done on all 100 bootstrapped splines, with the median of each measure reported. Hartigan & Hartigan’s dip test for unimodality was used to determine the continuity of AmFRET values within the determined transition range. Plots are classified as “discontinuous” or “continuous” for p-values less than or greater than 0.05, respectively.

Plasmids that were determined to have a continuous transition were further classified as uniformly low (“low”), uniformly high (“high”), or transitioned from low to high with increasing concentration (“low to high”). This was done by extracting the minimum and ending AmFRET values from the spline fit. As a reference for the AmFRET value that indicates the start of a high FRET state, the AmFRET value of the transition start point of a control plasmid, rhx0927, was used. For each plasmid, if its minimum and ending AmFRET values fell below the reference, it was

classified as low. If both fell above, it was classified as high. Those with a minimum value below and an ending value above were classified as “low to high”. The majority classification of replicates for each plasmid is reported.

Calculating centrality measures

To determine betweenness and degree of centrality, we extracted interactions involving DFD-containing proteins (listed in Table S1) from STRING version 12.0,¹²² considering only physical interactions with scores of 900 or higher. Using NetworkX v3.1, we analysed these interactions as an undirected graph to calculate betweenness and degree of centrality. For proteins with multiple DFDs, we classified a protein as discontinuous if any of its DFDs were identified as discontinuous.

Determination of positive nucleating interactions

We first excluded files with fewer than 2500 events positive for mEos3 or mean acceptor intensities less than 3.5 p.d.u. From this, any DFD or seed left with less than 25% of their original instances after filtering were removed from the analysis completely. Next, we identified nucleating interactions as DFD pairs that decreased the C50 and increased the fraction assembled (fgate). We standardized all variables for each experimental batch of DFDs to a mean of 0 and variance of 1. We then determined the outlier degree for C50 and fgate based on the number of interquartile ranges below or above the median for these values. This was done directionally on a per-DFD basis. We defined the “nucleating interactions” for a given mEos3-fused DFD as those whose mean of these two values (reported as “seedability”) is greater than or equal to 3 standard deviations above the mean of all seedability values. We confirmed that the seedability values for most DFD pairs partitioned with that of two negative controls included for each DFD.

To evaluate the reproducibility of our assay, we replicated it for a set of 36 DFDs. This replicate analysis mirrored the original, except it utilized the DFD distributions and cutoff values from the first set. The Pearson correlation (R) between the two sets was 0.91 ($p < 0.0001$). To minimize the impact of random variations in negatives and outliers, we excluded double negative instances, resulting in a slightly altered Pearson correlation (R) of 0.90 ($p < 0.0001$). Of the 3423 DFD + seed combinations reassessed, 16 showed inconsistent hit-calling, indicating an assay consistency rate of 99.53% with a 95% confidence interval ranging from 99.30% to 99.76%.

Approximating C_{sat} and supersaturability

To generate an average DAmFRET curve, we computed the mean of each histogram bin in the DAmFRET dataset, focusing on bins containing a minimum of 100 cells. The average DAmFRET curve for each DFD in the presence of its self-seed was fit to a Weibull function as follows. We first calculated the average AmFRET value in each concentration bin. The resulting curves resemble the fraction-assembled curves except that the asymptote is the maximum AmFRET value rather than 1. Therefore, we used the following equation for fitting:

$$f(c) = A \left[1 - \exp \left(- \left(\frac{c}{C_{50}} \right)^n \right) \right] \quad (\text{Eq. 1})$$

Here A is the AmFRET asymptotic value of the curve, c is the concentration, C_{50} is the concentration at which the curve, in the presence of its self-seed, has reached 50% of its asymptotic value, and n describes the steepness of the stretched exponential. Initial values of the parameters were chosen based on Gaussian smoothed versions of the curves and constrained in the fit to at minimum a 2-fold change from those initial guesses. The parameter was constrained between 0.1 and 10 based on expected reasonable values of that parameter. At low and high concentration values, the average AmFRET values are clearly unstable and influenced by noise and minor compensation errors. Therefore, we chose the beginning and ending points of each curve by visual inspection, choosing starting points where the curve begins to increase and ending points where the curve levels off. Error values were determined from Monte Carlo simulations as in the fitting of fraction assembly.

To approximate the supersaturability of DFDs, we used the ratio of the average C50 in the presence of the respective DFD's seed ($C50_{\text{seeded}}$) to the average C50 in the presence of a nullseed ($C50_{\text{stochastic}}$). Ratios were only calculated between C50s obtained from the same batch run.

Cell culture

HEK293T cells and THP-1 cells were purchased from ATCC. THP-1 PYCARD-KO (thp-koascz) cells were purchased from InvivoGen. HEK293T cells were grown in Dulbecco's Modified Eagle's Medium (DMEM) with L-glutamine, 10% fetal bovine serum (FBS), and PenStrep 100 U/mL. THP-1 cells were grown in Roswell Park Memorial Institute (RPMI) medium 1640 with L-glutamine and 10% FBS. All cells were grown at 37°C in a 5% CO₂ atmosphere incubator. Cell lines were regularly tested for mycoplasma using the Universal mycoplasma detection kit (ATCC, #30-1012K).

Generation of stable cell lines

Stable cell lines were created as described.¹² Briefly, constructs were packaged into lentivirus in a 10 cm plate 60% confluent of HEK293T cells using the TransIT-LT1 (Mirus Bio, MIR2300) transfection reagent and 7 µg of the vector, 7 µg psPAX2, and 1 µg pVSV-G. Lentivirus was harvested and incubated with 293T with polybrene or infected at 1000x g for 1 h for THP-1 cells. For transduction of pCW57.1 derived vectors, HEK293T and THP-1 cells were selected with Puromycin (1 µg/mL) for 7 days. After this time, cells were sorted for positive expression of mScarlet-I and expanded in continuing selection with puromycin. For transduction of plasmids encoding fusions to miRFP670nano-Cryclust, THP-1 and HEK293T cells were selected with hygromycin B (350 µg/mL and 150 µg/mL, respectively) for 7 days. Cells were sorted for positive expression of miRFP670nano and expanded for further experiments with continued selection. To generate THP-1 PYCARD-KO + FADD-KO cells, sgRNA targeting FADD exon1 was cloned into the lentiCRISPR v2-Blast (Addgene #83480). This vector was packaged into lentivirus as described above. THP-1 PYCARD-KO cells were transduced using spinfection and supplemented with polybrene. 24 h after spinfection, media was replaced. 48 h after spinfection, cells were selected with blasticidin (1 µg/mL). After 10 days of blasticidin selection, single-cell clonal expansion was done by serial dilution of resistant cells to achieve complete knockouts. Selected wells were analysed by immunoblot to confirm the absence of FADD protein and sequence-verified.

High-content imaging analysis

High-content imaging was performed on the Opera Phenix high-content screening system (PerkinElmer) using a 63x water immersion objective. Briefly, yeast (rhy2977) transformed with individual plasmids were cultured and induced as for DAMFRET assays. Then, 10 µL were transferred into a well containing 90 µL of SGal-URA of a 96-well optically clear flat-bottom plate (PerkinElmer 6055302). Data analysis of the high content imaging was performed in Fiji. Images of mEos3 were acquired using 488 nm excitation and a standard GFP filter set. Small zstacks were acquired over 5 µm total range with 1 µm steps. The image containing the brightest mEos3 signal was used. The mEos3 signal was then background-subtracted with a rolling ball radius of 100 pixels then found and converted to Fiji ROIs using the Fiji Default method of traditional image thresholding. The mean, standard deviation, and aspect ratio (AR) were measured for each object. The coefficient of variation (CV) in pixel intensity was calculated for every object following the formula: $CV = \text{Std Dev} / \text{Mean} * 100$. The wells were divided into 3 categories based on their AR and CV. Objects that had a CV > 55 and an AR > 1.159 were designated "fibrillar". Objects that had a CV > 55 and an AR < 1.16 were designated "punctate". Finally, objects that had a CV < 55 and an AR < 1.16 were designated "diffuse". These cutoffs were determined manually from a visual inspection of the data. Plasmids rhx4763 - rhx4767 were acquired in a second data set that used different laser powers and integration times. Hence, for these plasmids specifically, "fibrillar" had a CV > 17 and an AR > 1.4, "punctate" had a CV > 17 and an AR < 1.41, and "diffuse" had a CV < 18 and an AR < 1.41. These cutoffs were determined manually from a visual inspection of the data. The results were then manually verified for all wells. Seven plasmids were inconsistently classified by these cutoffs. Of these, manual inspection confirmed "diffuse" morphology for rhx1033 and rhx1133,

and prompted reclassification of rhx2935, rhx2637, rhx1121, rhx1070, and rhx1055 from “fibrillar” to “punctate”. Plasmid rhx2232 was incorrectly classified as punctate but images display fibrillar morphology. Three others (rhx1113, rhx1097, rhx2937) had anomalously high AR due to low expression. Representative microscopy images are included for all plasmids in our repository.

Fluorescence microscopy and optogenetic nucleation

The yeast and mammalian cells were imaged in an LSM 780 microscope with a 63x PlanApochromat (NA = 1.40) objective. T-Sapphire was excited with a 405 nm laser. mEos3 and mScarlet-I were excited with a 488 nm and 561 nm laser, respectively. For time-lapse imaging, samples were maintained at 37°C and 5% CO₂ with a stage top incubator. To stimulate Cry2clust we used the 488 nm laser at a power setting of 50% for a pulse of 10 s, which is the amount of time it took to scan the user-generated region of interest unless indicated otherwise. 561 and 633 nm lasers were used for imaging mScarlet-I and mRFP670nano, respectively. Pyroptosis events were tracked by incorporating the Sytox Orange reagent into the cell. To quantify the CV, images were subjected to an in-house Fiji adapted implementation of Cellpose for cellular segmentation.¹²³ The Cellpose-generated regions of interest (ROIs) were used to measure specified imaging channels.

For quantification of cell death events using IncuCyte (Sartorius), THP-1 cells were plated on a 24 or 96 well plate at a density of 4x10⁴/well or 1x10⁴/well, respectively, with PMA (10 ng/mL) for 16 h. Media was replaced with fresh media supplemented with Annexin V-Alexa488 and Sytox Orange (1:1000). For AIM2-Cry2clust activation, an initial collection of unexposed measurements was taken for 30 min. Then, the plate was exposed to 488 nm laser every 5 min. For treatments with poly(dA:dT), cells were treated and immediately subjected to imaging every 30 min for 19 h. Positive cells for either fluorophore were identified using the integrated software in the IncuCyte instrument.

For optogenetic activation of APAF1^{CARD} and NCLR4^{CARD}, HEK293T cells expressing lentivirus constructs were seeded on a 35-mm dish (ibidi) at a density of 4x10⁴/mL with 2 mL of media. The next day, dox was added at a concentration of 1 µg/mL to induce the expression of mScarlet-I tagged proteins. 24 h after protein induction, media was replaced with fresh media supplemented with Incucyte® Caspase-3/7 Dye (1:1000) 2 h prior to the experiment or Annexin V-Alexa488. Cells were imaged using a spinning-disk confocal microscope (Nikon, CSU-W1) with a ×60 Plan ApoChromat objective (NA = 1.40) and a Flash 4 sCMOS camera (Hamamatsu). A region of interest (ROI) was selected to induce optogenetic activation for indicated times using a 488 nm laser at 50% laser power for the indicated time ranging from a fraction of a second to 30 s. ROIs were generated by Cellpose segmentation algorithm around each cell contour. These ROIs were then used to measure the area, mean, standard deviation, and integrated density of each cell on the 488 nm and 560 nm fluorescence channels.

Protein immunodetection

We performed capillary based protein immunodetection (Wes, ProteinSimple) as described.¹² Briefly, protein lysates were prepared as per recommended manufacturer instructions to a final concentration of 1 µg/mL. An assay plate was filled with samples, blocking reagent, primary antibodies (1:50 dilution for anti-Actin, 1:200 dilution for anti PYCARD), HRP-conjugated secondary antibodies and chemiluminescent substrate. The plate was subjected to electrophoretic protein separation and immunodetection in the fully automated capillary system. The resulting data was processed using the open-source software Compass (<https://www.proteinsimple.com/compass/downloads/>) to extract the intensities for the peaks corresponding to the expected molecular weight of proteins of interest. For western blot, cells were centrifuged at 1000 x g for 5 min and resuspended in lysis buffer (50 mM Tris (pH 7.4), 137 mM NaCl, 1 mM EDTA, 1% Triton X-100, 10 mM DTT (dithiothreitol), cOmplete Protease Inhibitor (1 tablet / 10 mL) (Roche, 11697498001). Protein lysates were resolved on a NuPAGE™ 4 to 12%, Bis-Tris gel and transferred onto a PVDF membrane (IPVH00010, Millipore) using the Pierce Power Blotter (ThermoFisher). The membrane was blocked with 5% skim milk and incubated overnight with the antibodies: anti-Actin (1:1000, sc-8432), anti-FADD (1:500, 05-486) and anti-PYCARD

(1:1000, sc-514414). Primary antibody was removed by several washes with TBS + 0.1% Tween-20 and then subjected to incubation with secondary antibody (anti-mouse-HRP, 7076S, Cell Signaling Technology). The detection of protein bands was then carried out using enhanced chemiluminescence (ECL) (SuperSignal West Pico Chemiluminescent Substrate, ThermoFisher, 34577). The chemiluminescent signal was acquired by placing the membrane in a film cassette and exposing it to X-ray film (Kodak) at varying durations in a darkroom. After exposure, films were developed using an automatic film processor.

Semidenaturing Detergent-Agarose Gel Electrophoresis (SDD-AGE)

SDD-AGE was performed as previously described.¹⁶ Briefly, cells were lysed using a 2010 Geno/Grinder with bead-beating. Samples were prepared with 2% sarkosyl and separated in a 1.5% agarose gel with 0.1% SDS. The distribution of mEos3-fused proteins was analysed directly in the gel with a GE Typhoon Imaging System. Images were processed to remove background using a 250-pixel rolling ball, cropped, and contrast-adjusted.

Quantification and statistical analysis

Two-sided Student's t-tests were used for significance testing unless stated otherwise for two sample comparisons. The graphs represent the means $\pm$ SEM of independent biological experiments unless stated otherwise. Statistical analysis was performed using GraphPad Prism 9, Python and R packages.

Supplementary Materials

Experiments to investigate the nature of DFD assembly

DAmFRET data do not clarify if self-assembly involves native DFD interactions rather than amyloid-like misfolding.¹⁶ To address this question, we introduced point mutations to disrupt assembly via conserved known interfaces between folded DFD subunits.^{23,37} Across the multiple DFDs examined, all such mutations indeed reduced or eliminated the high-AmFRET population (Figure S2A). To directly evaluate the nature of DFD assembly in our experiments, we subjected the seeded and unseeded cells to semi-denaturing detergent-agarose gel electrophoresis (SDDAGE), a technique that distinguishes amyloids from other protein states based on their detergentresistance and size dispersity.¹²⁴ We found that unlike our amyloid control (RIPK1^{RHIM}), none of the DFD assemblies survived sarkosyl exposure (Figure S2B), consistent with their retaining the death fold rather than misfolding into amyloid. Ongoing work is elucidating the physical basis of these nucleation barriers.

Acknowledgements

We thank Nick Grishin and Lisa Kinch for creating structure-guided alignments of DFDs early in this work, and Mark Miller for assistance with illustrations. This work was performed to fulfill, in part, requirements for ARG's thesis research in the Graduate School of the Stowers Institute for Medical Research. This work was supported by the National Institute of General Medical Sciences (Award Number R01GM130927, to RH) and the National Institute on Aging (Award Number F99AG068511, to ARG) of the National Institutes of Health, the American Cancer Society (RSG-19-217-01-CCG to RH), and the Stowers Institute for Medical Research. The funders had no role in study design, data collection and analysis, or manuscript preparation. The content is solely the responsibility of the authors and does not necessarily represent the official views of the funders.

Additional information

Author Contributions

Conceptualization, A.R.G., S.V., and R.H.; Methodology, A.R.G., T.M., S.V., J.J.L., J.R.U. and R.H.; Formal Analysis, A.R.G., T.M., J.J.L. and J.R.U.; Investigation, A.R.G., T.M., S.V., J.J.L., J.W., X.S., and D.B.; Writing – Original Draft, A.R.G. and R.H.; Writing – Review & Editing, A.R.G., S.V., T.M. and R.H.; Funding Acquisition, A.R.G. and R.H.; Supervision, R.H.

Data availability

All DAMFRET and representative microscopy data can be accessed interactively on a dedicated website https://simrcompbio.shinyapps.io/HalfmannLab-Nucleating_Interactome/. Original data underlying this manuscript can be accessed from the Stowers Original Data Repository at <http://www.stowers.org/research/publications/libpb-2387>

Funding

Funder	Grant reference number	Author
HHS NIH National Institute of General Medical Sciences (NIGMS)	R01GM130927	Randal Halfmann
HHS NIH National Institute on Aging (NIA)	F99AG068511	Alejandro Rodriguez Gama
American Cancer Society (ACS)	RSG-19-217-01-CCG	Randal Halfmann
Stowers Institute for Medical Research (SIMR)		Randal Halfmann

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Additional files

Supplemental Figure 1. Sequence, imaging and DAMFRET analysis reveal diverse sequence-encoded phase behaviours of DFDs, related to Figure 1. A. Schematic diagram of all DFDs characterized in this paper, and their classification into structural subfamilies. Tandem DFDs are highlighted in red, but were analysed with their corresponding single DFD subfamilies. B. Matrices of predicted alignment error (PAE) for the indicated regions of proteins containing two DFDs, as reported in the AlphaFold Protein Structure Database¹²⁰, grouped into two categories according to interdomain PAE values consistent with either independent (left) or dependent (right) relative geometries of the DFDs. C. DAMFRET profiles of representative DFDs classified as continuous or discontinuous. D. Classification of continuous DFD as “low”, “low to high”, or “high” by thresholding on the minimum and ending AmFRET values of a fitted spline, normalized to that of a control DFD. E. Classification of the proteins as entirely diffuse, fibrillar or punctate based on boundaries on the scatter plot of coefficient of variation vs aspect ratio. The colored circles represent the mean and covariance of the values for each category. F. Images of yeast expressing representative DFDs, in the absence of seeds, classified as fibrillar that produced continuous (low to high) DAMFRET profiles. Scale bar 10 µm. G. Images of yeast expressing representative DFDs, in the absence of seeds, classified as fibrillar that produced discontinuous DAMFRET profiles. Scale bar 10 µm. H. Images of yeast expressing representative DFDs, in the absence of seeds, classified as punctate that produced continuous (low to high) DAMFRET profiles. Scale bar 10 µm. I. Representative confocal microscopy images of yeast expressing the indicated DFD constructs in the presence of ASC^{FL} or CARD14^{CARD} seeds. The images show the emergence of filaments only from µNS seeds containing the matching DFD.

Supplemental Figure 2. Self-assembly involves subunit interfaces shared with solved DFD polymer structures, related to Figure 1. A. Representative DAMFRET plots for the indicated DFDs with the indicated point mutations. The horizontal line approximates the mean AmFRET value for

monomeric mEos3. Procedure defined units (p.d.u.). B. Image of SDD-AGE showing the size distribution of detergent-resistant multimers (where present) of mEos3-fused proteins expressed in yeast. The amyloid-forming protein, RIPK1^{RHIM}, formed detergent-resistant multimers whereas all DFD multimers were detergent-labile.

Supplemental Figure 3. Proteins with DFDs that have seedable and/or discontinuous DAmFRET are central to their physical interaction networks and are more likely to be supersaturated in vivo, related to Figure 2. A. Transcripts encoding proteins with discontinuous DFDs have higher expression in immune cells. P values are from Mann-Whitney test (see also Table S5). Transcripts per million (TPM) values are from the immune cell data of the Human Protein Atlas, comprising 18 cell types and total Peripheral Blood Mononuclear Cells (PBMC). B. Heatmap of protein abundance relative to reference, of discontinuous and continuous DFD containing proteins for the indicated tissues. Tissues are ordered by significance. P values are from Mann-Whitney test (see also Table S5). Protein abundance values are from the Proteome Map of the Human Body³⁰. C. Bar plot of the Spearman R correlation between immune cell type transcript abundance and C_{sat} values of DFDs shows consistent negative and significant anticorrelation among immune cell types. Data were obtained from the immune cell section of the Human Protein Atlas. D. Boxplot comparing the betweenness (left) and degree centrality (right) of DFDcontaining proteins that are either non-seedable or continuous (n = 37) to those that are both seedable and discontinuous (n = 14). Seedable, discontinuous proteins were found to have a significantly higher betweenness and degree centrality than non-seedable or continuous proteins. Mann-Whitney U = 145.5 (p = 0.012) and U = 146.5 (p = 0.017), respectively. E. Permutation test (n = 10000) comparing betweenness centrality (top) and degree centrality (bottom) for continuous and discontinuous DFDs. Mann-Whitney U = 274 (p = 0.028) and 242 (p = 0.008), respectively. F. Boxplot comparing the betweenness (left) and degree centrality of Sterile Alpha Motif (SAM) containing proteins network with continuous and discontinuous DFD networks. SAM containing protein network was extracted from STRING version 12.0 physical interaction database with a score higher than 900. Mann-Whitney U = 34 (p < 0.001) and U = 145 (p = 0.799), respectively between SAM (n=17) and discontinuous DFD (n=18).

Supplemental Figure 4. Proteins characterized as signaling adaptors display discontinuity in its DFD and FL context, related to Figure 2. A. Pairs of DAmFRET plots comparing the behaviours of representative DFDs and their corresponding FL proteins. Dashed horizontal lines approximate the mean AmFRET value for monomeric mEos3. FL MAVS has appreciable AmFRET in the supersaturated state that we attribute to its mitochondrial localization signal. B. Left, DAmFRET plot of ASC^{PYD} expressed alone. Right, DAmFRET plot of ASC^{PYD} coexpressed with FL NLRP3 showing persistence of the supersaturated bottom population indicating that FL NLRP3 oligomers are not active (in the absence of stimulation).

Supplemental Figure 5. Full length proteins displaying discontinuous profiles can be self-seeded, related to Figures 1 and 2. A. Representative DAmFRET plots for the indicated full length proteins. Seeds comprised μ NS-mCardinal fusion proteins either lacking (left) or containing (right) the corresponding DFD. The horizontal dotted lines approximate the mean AmFRET value for monomeric mEos3.

Supplemental Figure 6. Characterization of engineered THP-1 cell lines and apoptosome assembly, and correlation of DFD supersaturation with cell mortality in the human body, related to Figures 3 and 4. A. DAmFRET plots of APAF1^{CARD} and CASP9^{CARD} measured in the presence of the indicated “seeds” expressed in trans. Both proteins fail to populate a high-AmFRET state. B. Western blot verifying the knock-out status of ASC and/or FADD in the respective engineered stable THP-1 cell lines. Actin is the loading control. C. Cartoon depicting the doxycycline-inducible ASC-mScarlet-I that replaced endogenous ASC in THP-1 PYCARD-KO cells. D. Representative capillary western blots comparing expression levels of the dox-inducible ASC-mScarlet-I construct alongside endogenous ASC. Actin is the loading control. E. Quantification of the data showing significantly lower-than endogenous levels of ASC in the engineered construct at even the highest level of induction by doxycycline. For each condition, one million cells were sorted and lysed. F. Scatter plot showing the relationship between ASC supersaturation (as approximated by the ratio

of transcription levels and C_{sat} values) and mean lifespan for each cell type as indicated in Figure 4H. The red line represents the best-fit power-law regression, obtained by performing linear regression in log-log space. The shaded region represents the 95% confidence interval for the trend line. Spearman $R = -0.87$ (two-tailed $p = 0.0000057$). G. Bar plot of the Spearman R correlation between supersaturation and cell mean lifespan of cell types as shown in Figure 4H for each DFD. Transcript levels for each cell type were obtained from the single cell RNA dataset of the Human Protein Atlas.

Supplemental Figure 7.  Generation and validation of the DFD nucleating interactome, related to Figure 5. A. Illustration of how the library of all pairs of DFDs was created. An arrayed sublibrary of yeast transformed with 105 DFD-mEos3 fusions was mated to a separate arrayed sublibrary of yeast strains expressing 107 chromosomally integrated DFD- μ NSmCardinal fusions, to create a library of 12,660 diploid strains representing all pairwise combinations. B. Left, DAmFRET was run on all pairwise combinations. Only high quality datasets – having a total cell count greater or equal to 2500 and a mean acceptor intensity greater or equal to 3.5 p.d.u. – were used in the analysis. Right, DAmFRET plots of FADD^{FL} either null-seeded (lacking a DFD) or self-seeded. Nucleating interactions (as shown by the self-seeded example) are indicated by a reduced C50 and increased percentage of cells with self-assemblies (those above the gate delimiting low FRET, shown in orange). C. Hits are determined by a multiparameter combination of the degree of C50 outlier and the degree of fraction assembled outlier as defined by how many interquartile ranges (IQR) a plot is below or above the median, respectively. Points on the graph are shaded by this parameter. The leftmost set of boxplots show the distribution of standardized log₁₀ C50 and fraction assembled for all seeds for a representative protein, FADD^{FL}. The middle boxplot shows the average outlier degree value of these two parameters. This value is referred to as “seedability” throughout the text and is used in determining hits. The cutoff value for hit determination was set to be 3 standard deviations above the mean of all seedability values across the screen. The scatter plot on the right shows the standardized log₁₀ C50 and fraction assembled values, depicting the contribution of both to the scoring value and positive nucleating interactions within the green box. D. Top, scatter plot of seedability values from the two replicate experiments containing 3478 DFD + seed combinations. Points are colored according to agreement between the two experiments. Gray rectangles indicate seedability values of negative interactions, with the dark gray square containing the interaction found to be negative for both instances. Bottom, the replicates after the removal of instances found to be negative in both experiments to reduce random and outlier effects. The line of best fit is shown in red. Pearson correlation R between the two experiments = 0.91 ($p < 0.0001$). In order to reduce the effects of random variations of the negatives as well as outlier effects, we omitted double negative instances. In this case, the Pearson correlation $R = 0.90$ ($p < 0.0001$). E. Seventeen DFD + seed combinations that had inconsistent hit-calling out of the 3478 combinations reassessed. From this we determine that our assay had a consistency of 99.51% with a 95% confidence interval of 99.28 - 99.74%. F. Bar plots showing the number of hits per experiment, as well as total number of hits, determined for each replicated DFD. Blue and yellow indicate separate hit counts for each experiment. Gray are the number of unique hits found across both experiments. What percentage of seeds are consistently called hits are shown for each DFD. The left bar plot shows the overall summary of all DFDs included in both sets.

Supplemental Figure 8.  Demonstration of conserved energy storage capacity of DFDs, related to Figure 6. A. DAmFRET data of DFD-only and full length inferred DISC components from the model sponge, *Amphimedon queenslandica*. The distant homolog to human FADD exhibits supersaturability in its FL and isolated DFDs. B. DAmFRET data of DFD-only and full length inflammasome components from the model fish *Danio rerio*.

Supplemental Table 1.  Metadata, construct specifics, and data for the human DFDs analysed in this study, related to Figures 1 and 2. ND (not determined). NA (not applicable/available). FL (fulllength). ^arhx1073b, rhx1102b, rhx1108, and rhx1135 did not have a majority automated classification and were therefore classified by manual inspection. ^bInspection of the DAmFRET plots of the six nonseedable discontinuous DFDs reveal that most are only slightly discontinuous,

and include CARD9^{CARD} which we previously showed to have a negligible nucleation barrier³⁷, indicating that at least some of these six are misclassified due to heterogeneity between cells around C_{sat} .

Supplemental Table 2. [↗](#) Metadata, construct specifics, and data for additional proteins analysed in this study, related to Figure 2. FL (full-length). ^aMutations were made to inactivate enzymatic activity for cell death effectors.

Supplemental Table 3. [↗](#) Specifics of lentivirus vectors used in this study, related to Figures 4 and 5.

Supplemental Table 4. [↗](#) Nucleating interactome of DFDs, related to Figure 5. Note that plasmids rhx2938, rhx1071, rhx2934, rhx1113, rhx1064, rhx2933, and rhx1368; and seeds from rhx2938, rhx2933 and rhx2934; did not pass quality control as outlined in methods and were therefore omitted from the analysis.

Supplemental Table 5. [↗](#) Transcript and protein abundance statistics, related to Figure S3.

Supplemental Movie 1. [↗](#) Time-lapse movie of cells expressing opto-APAF1 (top) and CASP9-mScarlet-I (bottom) showing one minute of blue light induction and subsequent recovery, related to Figure 3. Non-supersaturable opto-APAF1 allows clusters to dissolve following stimulation. Scaler bar 10 μ m.

Supplemental Movie 2. [↗](#) Time-lapse movie of cells expressing opto-NLRC4 (top) and CASP9(CASP1^{CARD})-mScarlet-I (bottom), showing one minute of blue light induction and subsequent recovery, related to Figure 3. Supersaturable opto-NLRC4 continues to drive cluster growth following stimulation. Scaler bar 10 μ m.

Supplemental Movie 3. [↗](#) Time-lapse movie of cells expressing opto-AIM2, following 10 seconds of blue light induction, related to Figure 4. Left panel: opto-AIM2. Middle panel: Sytox Orange fluorescence. Right panel: Bright field. Following 10 seconds of blue light induction, opto-AIM2 clusters form, leading to cell death (Sytox Orange uptake). Scale bar: 10 μ m.

Supplemental Movie 4. [↗](#) Time-lapse movie of cells expressing opto-AIM2(F27G), following 10 seconds of blue light induction, related to Figure 4. Left panel: opto-AIM2(F27G). Middle panel: Sytox Orange fluorescence. Right panel: Bright field. Following 10 seconds of blue light induction, opto-AIM2(F27G) clusters form and rapidly dissociate, failing to induce cell death. Scale bar: 10 μ m.

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Peer reviews

Reviewer #1 (Public review):

This is a high-quality and extensive study that reveals differences in the self-assembly properties of the full set of 109 human death fold domains (DFDs). Distributed amphifluoric FRET (DAmFRET) is a powerful tool that is applied here for a comprehensive examination of the self-assembly behaviour of the DFDs, in non-seeded and seeded contexts, and allows comparison of the nature and extent of self-assembly. The work reveals the nature of the barriers to nucleation in the transition from low to high AmFRET. Alongside analysis of the saturation concentration and protein concentration in the absence of seed, the work demonstrates that the subset of proteins that exhibit discontinuous transitions to higher-order assemblies are expressed more abundantly than DFDs that exhibit continuous transitions. The experiments probing the ~20% of DFDs that exhibit discontinuous transition to polymeric form suggest that they populate a metastable, supersaturated form, in the absence of cognate signal. This is suggestive of a high intrinsic barrier to nucleation.

The differences in self-assembly behaviour are significant and highlight mechanistic differences across this large family of signalling adapter domains, with identification of a small number of key supersaturated adapters, which exhibit higher centrality within networks, and can amplify signals and transduce them to effectors as required. The description of some supersaturated DFD adaptors as long-term, high-energy storage forms or phase change adaptors is attractive and is a framework that addresses many of the requirements for on-demand signaling and amplification in innate immunity. The identification of only a small number of key adaptors and high specificity suggests a mechanism for insulation of pathways from each other and minimisation of aberrant lethal consequences.

An optogenetic approach is applied to initiate self-assembly of CASP1 and CASP9 DFDs, as a model for apoptosome initiation in these two DFDs with differing continuous or discontinuous assembly properties. This comparison reveals clear differences in the stability and reversibility of the assemblies, supporting the authors' hypothesis that supersaturation-mediated DFD assembly underlies signal amplification in at least some of the DFDs. The study also reveals interesting correlations between supersaturation of DFD adaptors in short- and long-lived cells, suggestive of a relationship between mechanism of assembly and cellular context. Additionally, the interactions are almost all homomeric or limited to members of the same DFD subfamily or interaction network and examination of bacterial proteins from innate immunity operons suggest that their polymerisation could be driven by similar mechanisms. Future detailed studies that probe the roles and activities of DFDs identified with continuous or discontinuous barriers to nucleation, through mutational analysis, in chimeric proteins and with high resolution studies of the assemblies, can build on this methodology and database.

The Discussion effectively places this work in the context of innate immunity effectors and adapters, explains and provides a justification of the phase change material analogy, and contrasts this mechanism with phase separation. The breadth and depth of the experimental investigations allow a new view of the role of nucleation barriers and supersaturation in DFD assembly and innate immunity pathways.

<https://doi.org/10.7554/eLife.107962.2.sa2>

Reviewer #2 (Public review):

This work studies the self-association behavior of 109 human Death Fold Domains (DFD) in eukaryotic cells and its connection to their function in innate immune signalosomes.

Using an amphifluoric FRET (DAmFRET) method previously developed by the authors, self-association is monitored as a function of protein concentration by Förster Resonance Energy Transfer in the cell.

Several DFDs are found to be in a supersaturable state and are considered energy reservoirs necessary for signal amplification.

The revised manuscript addresses most of the original concerns, resulting in a significant improvement.

The following observations are made:

- (1) A group of DFDs shows a bimodal FRET distribution of no FRET and high FRET values at low and high protein concentration, which indicates a nucleation barrier. This conclusion is corroborated by the modification from a discontinuous to a continuous FRET transition by expressing a structural template or seed. The authors find that DFDs displaying discontinuous FRET behavior are supersaturated, and those that retain their discontinuous behavior in the context of the full-length protein correspond to protein adaptors of innate immune signalosomes.
- (2) The authors indicate that the adaptors of inflammatory signalosomes act as energy reservoirs for signal amplification. This is not demonstrated, but it is assumed that the energy stored in the supersaturated state is released upon polymerization.
- (3) This work also includes evidence showing that nonsupersaturable and supersaturable constructs of caspase-9 form puncta that dissolve or persist, respectively, upon apoptosome stimulation. The supersaturable construct also induces massive cell death, in contrast to the nonsupersaturable form. Although not demonstrated, these results could be related to the level of signal amplification.
- (4) The cell's lifespan depends on the supersaturation levels of certain DFDs.
- (5) Polymerization nucleated by interaction between DFDs from different pathways (different signalosomes) is rare.
- (6) The study demonstrates the presence of nucleation barriers, inferred from supersaturable conditions, in the adaptor orthologs of zebrafish (*Danio rerio*) and the model sponge *Amphimedon queenslandica*, which indicates that this characteristic is conserved.

<https://doi.org/10.7554/eLife.107962.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Both reviewers indicated broad approval of the revised work, for which we are grateful.

Reviewer #1 requested no further changes.

Reviewer #2's Public review states:

The authors indicate that the adaptors of inflammatory signalosomes act as energy reservoirs for signal amplification. This is not demonstrated, but it is assumed that the energy stored in the supersaturated state is released upon polymerization.

The “assumed” link between supersaturation and energy release is in fact a thermodynamic necessity. Supersaturation is, by definition, a high free energy state. Our data shows that triggering nucleation via optogenetics results in an immediate avalanche of polymerization and cell death. This is not an assumption; it is a direct observation of work performed by the system when the kinetic barrier is removed.

Reviewer #2 recommended:

Ideally, signal amplification could be tested by determining the levels of the final product, e.g., cytokines, activated caspases...

We did measure CASP3/7 activation, demonstrating a correlation with supersaturation of upstream adaptors. We do agree however that measuring the levels of other signaling products, including for each of the supersaturated pathways, would strengthen our claims. This will be the subject of future work.

The authors indicate a significant anticorrelation between the saturating concentrations and the transcript abundances (Figure 2B), reporting an $R = -0.285$.

This is correct... no change appears to be requested or warranted.

The following is the authors’ response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This is a high-quality and extensive study that reveals differences in the self-assembly properties of the full set of 109 human death fold domains (DFDs). Distributed amphifluoric FRET (DAmFRET) is a powerful tool that reveals the self-assembly behaviour of the DFDs, in non-seeded and seeded contexts, and allows comparison of the nature and extent of self-assembly. The nature of the barriers to nucleation is revealed in the transition from low to high AmFRET. Alongside analysis of the saturation concentration and protein concentration in the absence of seed, the subset of proteins that exhibited discontinuous transitions to higher-order assemblies was observed to have higher concentrations than DFDs that exhibited continuous transitions. The experiments probing the ~20% of DFDs that exhibit discontinuous transition to polymeric form suggest that they populate a metastable, supersaturated form in the absence of cognate signal. This is suggestive of a high intrinsic barrier to nucleation.

Strengths:

The differences in self-assembly behaviour are significant and likely identify mechanistic differences across this large family of signalling adapter domains. The work is of high quality, and the evidence for a range of behaviours is strong. This is an important and useful starting point since the different assembly mechanisms point towards specific cellular roles. However, understanding the molecular basis for these differences will require further analysis.

An impressive optogenetic approach was engineered and applied to initiate self-assembly of CASP1 and CASP9 DFDs, as a model for apoptosome initiation in these two DFDs with differing continuous or discontinuous assembly properties. This comparison revealed clear differences in the stability and reversibility of the assemblies, supporting the hypothesis that supersaturation-mediated DFD assembly underlies signal amplification in at least some of the DFDs.

The study reveals interesting correlations between supersaturation of DFD adapters in short- and long-lived cells, suggestive of a relationship between the mechanism of assembly and cellular context. Additionally, the comprehensive nature of the study provides strong evidence that the interactions are almost all homomeric or limited to members of the same DFD subfamily or interaction network. Similar approaches with bacterial proteins from innate immunity operons suggest that their polymerisation may be driven by similar mechanisms.

Weaknesses:

Only a limited investigation of assembly morphology was conducted by microscopy. There was a tendency for discontinuous structures to form fibrillar structures and continuous to populate diffuse or punctate structures, but there was overlap across all categories, which is not fully explored.

We agree that an in-depth exploration of aggregate morphology would be interesting, but we feel it has limited relevance to the central findings of the manuscript. Our analysis established a relationship between discontinuous transitions and ordering based on the assumption that ordered assembly by DFDs involves polymerization, for which there is much precedent in the literature. Nevertheless, polymers of similar structure can form with different kinetics and hence, polymerization does not by itself imply an ability to supersaturate. We see this empirically in the “fibrillar” column in Fig. 1B. We have now elaborated this important point more fully in the relevant results section and in the discussion. Only five of the 108 DFDs in Fig. 1B warrant additional explanation. CASP4^{CARD} and IFIH1^{tCARD} lacked AmFRET but formed puncta; this could result from interactions with endogenous structures or condensates. DAPK1^{DD} and UNC5A^{DD} were classified as continuous (low) and fibrillar, but their AmFRET values are in fact higher than monomer control revealing that the fibrils simply comprise a small fraction of the protein. The puncta of UNC5A^{DD} additionally do not resemble the fibrillar puncta of other DFDs; we suspect it may be a false-positive resulting from localization to mitochondrial or other intracellular membranes. Finally, CASP2^{CARD} was inadvertently classified as punctate; this turns out to have been a technical artifact that has now been corrected (the fibrils wrapped around the cell perimeter to form ring-like puncta with anomalously low aspect ratios). We have now updated the methods section describing manual validation of our automated classification procedure, including which samples required reclassification. We have also now included all microscopy data in the public repository accompanying this manuscript.

The methodology used to probe oligomeric assembly and stability (SDD-AGE) does not justify the conclusions drawn regarding stability and native structure within the assemblies.

The reviewer is correct that SDD-AGE does not provide evidence against non-amyloid misfolding. It merely provides evidence that the DFDs are not forming amyloid (which are characteristically sarkosyl resistant). We have revised the sentence and further clarified that the distinction with amyloid specifically is important because amyloid is the only known form of ordered assembly (other than DFD polymers) with a nucleation barrier large enough to support deep supersaturation. Together with the series of interfacial mutants tested (and shown to impede assembly in all cases), the lack of sarkosyl-resistance provides evidence that the discontinuous DFDs are assembling through canonical DFD subunit interfaces.

The work identifies important differences between DFDs and clearly different patterns of association. However, most of the detailed analysis is of the DFDs that exhibit a discontinuous transition, and important questions remain about the majority of other DFDs and why some assemblies should be reversible and others not, and about the nature of signalling arising from a continuous transition to polymeric form.

We focused on discontinuous DFDs because this property allows for executive control over their respective pathways. They make signaling switch-like, which we argue is essential for innate immune responses. By contrast, and as illustrated in Figure 6D, supersaturation is required for a DFD to drive its own polymerization – hence activation for a continuous DFD must be stoichiometrically coupled either with D/PAMP binding or positive feedback from downstream or orthogonal processes. We consider the principles underlying such regulation of signaling to be better established and understood than supersaturation, and hence built our narrative for this manuscript around the latter. Our original text addresses the fact that only a small fraction of DFDs are discontinuous. Specifically, this is expected in light of the fact that a) only one supersaturated DFD is needed to make a signaling pathway switch-like, and b) every supersaturated DFD renders the cell susceptible to spontaneous death. Evolution should therefore limit supersaturation to only the highly connected DFDs (i.e. adaptors), which is what is seen. In this view, the many nonsupersaturable DFDs have evolved to accessorize the central supersaturable DFDs with various sensor and effector modules. Our revised text attempts to further clarify this perspective.

Some key examples of well-studied DFDs, such as MyD88 and RIPK1, deserve more discussion, since they display somewhat surprising results. More detailed exploration of these candidates, where much is known about their structures and the nature of the assemblies from other work, could substantiate the conclusions here and transform some of the conclusions from speculative to convincing.

We were likewise initially surprised about the inability of MyD88 and RIPK1 to supersaturate. We have now elaborated in the Discussion how our findings can be rationalized by the apparent supersaturability of other adaptors in MyD88 and RIPK1 signaling pathways. We additionally discuss prior evidence that MyD88 may indeed be supersaturable, and how our experimental system could have led to a false positive in the unique case of MyD88.

The study concludes with general statements about the relationship between stochastic nucleation and mortality, which provide food for thought and discussion but which, as they concede, are highly speculative. The analogies that are drawn with batteries and privatisation will likely not be clearly understood by all readers. The authors do not discuss limitations of the study or elaborate on further experiments that could interrogate the model.

We have now added to the discussion a section on the limitations of our study. We appreciate that our use of “privatisation” was confusing and have omitted it. However, we consider the battery analogy to accurately convey the newfound function of DFDs and anticipate that this analogy will ultimately prove valuable for biologists. To facilitate comprehension, we have now broadened our description of phase change batteries in the introduction.

Reviewer #2 (Public review):

Summary:

The manuscript from Rodriguez Gama et al. proposes several interesting conclusions based on different oligomerization properties of Death-Fold Domains (DFDs) in cells, their natural abundance, and supersaturation properties. These ideas are:

- (1) DFDs broadly store the cell's energy by remaining in a supersaturated state;*
- (2) Cells are constantly in a vulnerable state that could lead to cell death;*
- (3) The cell's lifespan depends on the supersaturation levels of certain DFDs.*

Overall, the evidence supporting these claims is not completely solid. Some concerns were noted.

Strengths:

Systematic analysis of DFD self-assembly and its relationship with protein abundance, supersaturation, cell longevity, and evolution.

Weaknesses:

(1) On page 2, it is stated, "Nucleation barriers increase with the entropic cost of assembly. Assemblies with large barriers, therefore, tend to be more ordered than those without. Ordered assembly often manifests as long filaments in cells," as a way to explain the observed results that DFDs assemblies that transitioned discontinuously form fibrils, whereas those that transitioned continuously (low-to-high) formed spherical or amorphous puncta. It is unlikely to be able to differentiate between amorphous and structured puncta by conventional confocal microscopy. Some DFDs self-assemble into structured puncta formed by intertwined fibrils. Such fibril nets are more structured and thus should be associated with a higher entropic cost. Therefore, the results in Figure 1B do not seem to agree with the reasoning described.

The formation of microscopically visible elongated structures necessitates ordering on the length scale of 100s of nanometers. Otherwise surface tension would favor rounded aggregates. Conventional confocal microscopy is in fact well-suited and widely used to distinguish ordered from disordered assemblies in cells based on this principle.^{1,2} We are unaware of any examples of isolated DFDs forming regular polymers that manifest as round puncta or nets. The reviewer may be referring to full-length ASC, which forms a roughly spherical mesh of filaments because it has two DFDs joined by a flexible linker. This is not applicable to our analysis with single DFDs. Single DFDs polymerize in effectively one dimension; hence a spherical punctum formed by a single DFD can only happen through noncanonical interactions or clustering of small filaments, both of which reduce order relative to long filaments.

(2) Errors for the data shown in Figure 1B would have been very useful to determine whether the population differences between diffuse, punctate, and fibrillar for the continuous (low-to-high) transition are meaningful.

We have now performed two statistical analyses to address this. First, using Fisher's exact test, we observe a highly significant association between the DAMFRET and morphology classifications (p -value: 0.0001). Second, to specifically address whether the continuous (low to high) category has a preferred morphology, we applied an Exact Multinomial Test using the total frequencies of each morphology. This test revealed that all categories are significantly enriched for particular morphologies, as now indicated in the figure and legend.

(3) A main concern in the data shown in Figure 1B and F is that the number of counts for discontinuous compared to continuous is small. Thus, the significance of the results is difficult to evaluate in the context of the broad function of DFDs as batteries, as stated at the beginning of the manuscript.

Fig. 1B simply reports the numerical intersections between fluorescence distribution classifications and DAMFRET classifications. In Fig. 1F, our use of the chi-square test is justified by a sufficiently large sample size. Nevertheless, we obtain similar results with Fisher's exact test that accounts for smaller sample size (Odds Ratio: 75.0, P -value: < 0.0001). See also our response to the related critique by Reviewer 1 regarding the small number of discontinuous DFDs.

(4) The proteins or domains that are self-seeded (Figure 1F) should be listed such that the reader has a better understanding of whether domains or full-length proteins are considered, whether other domains have an effect on self-seeding (which is not discussed), and whether there is repetition.

We define and consistently use “DFDs” to refer to domains, and “FL” or “DFD-containing protein” to refer to FL proteins. The Figure 1 title and corresponding section title both indicate the data refer to “DFDs”. The text callout for Figure 1F also directs readers to Table S1 where we believe the self-seeding results and details of constructs are clearly presented. There is no repetition. We have modified the legend to clarify that “Each DFD was co-expressed with an orthogonally fluorescent μ NS-fused version of the same DFD.” We did not systematically evaluate seeding of FL proteins. We did however previously test self-seeding on seven representative FL proteins, and have now included those data in a new supplemental figure (S5). In short, only FL proteins with discontinuous distributions are self-seedable. These are limited to adaptors that had discontinuous seedable DFDs, revealing no adverse effect of FL protein context on seedability of adaptors (unlike receptors and effectors).

(5) The authors indicate an anticorrelation between transcript abundance and Csat based on the data shown in Figure 2B; however, the data are scattered. It is not clear why an anticorrelation is inferred.

An anticorrelation is indicated by the clearly placed negative R value at the top of the graph and the figure legend describing the statistical analysis.

(6) It would be useful to indicate the expected range of degree centrality. The differences observed are very small. This is specifically the case for the BC values. The lack of context and the small differences cast doubts on their significance. It would be beneficial to describe these data in the context of the centrality values of other proteins.

The possible range of centrality scores is 0 - 1, where 1 represents a protein interacting with every other protein in the network (degree centrality) or is on the shortest path between every other pair of proteins in the network (betweenness centrality). The expected range is difficult to address, as centrality values strongly depend on the size and function of the network. We considered that the SAM domain network could provide the most relevant comparison to the DFD network, as SAM domains resemble DFDs in size and structure, function heavily in signaling, are comparably numerous (76 in humans), and many of them form homopolymers (but importantly of a geometry that does not support nucleation barriers). We found that SAM domains have much lower betweenness centrality in their physical interaction network as compared to discontinuous DFDs ($p = 0.0003$) while their degree centrality is not significantly different (Figure S3F). Nevertheless, we stress that what matters for our conclusion is that the continuous and discontinuous values are significantly different among DFDs. Since there is a large overlap in the distributions of centrality scores between the two classes of DFDs, we performed a more robust permutation test with the Mann Whitney U statistic and $n = 10000$. These tests reiterated that continuous and discontinuous DFDs have significantly different centrality scores (Degree centrality $p = 0.008$; Betweenness centrality $p = 0.028$) (Figure S3E).

(7) Page 3 section title: "Nucleation barriers are a characteristic feature of inflammatory signalosome adaptors." This title seems to contradict the results shown in Figure 2D, where full-length CARD9 and CARD11 are classified as sensors, but it has been reported that they are adaptor proteins with key roles in the inflammatory response. Please see the following references as examples: The adaptor protein CARD9 is essential for the activation of myeloid cells through ITAM-associated and Toll-like receptors. Nat Immunol 8, 619-629 (2007), and Mechanisms of Regulated and Dysregulated CARD11 Signaling in

Adaptive Immunity and Disease. Front Immunol. 2018 Sep 19;9:2105. However, both CARD9 and CARD11 show discontinuous to continuous behavior for the individual DFDs versus full-length proteins, respectively, in contrast to the results obtained for ASC, FADD, etc.

We rigorously counter the inconsistent usage of the term “adaptor” in the signalosome literature by quantifying the centrality of each protein in the physical interaction network of DFD proteins. Such analysis shows that BCL10, which is also described as an adaptor, is the more central member of the CARD9 and CARD11 (CBM signalosome) pathways, and is therefore more “adaptor-like”. We have now elaborated this view in the text.

FADD plays a key role in apoptosis but shows the same behavior as BCL10 and ASC. However, the manuscript indicates that this behavior is characteristic of inflammatory signalosomes. What is the explanation for adaptor proteins behaving in different ways? This casts doubts about the possibility of deriving general conclusions on the significance of these observations, or the subtitles in the results section seem to be oversimplifications.

We agree that our initial presentation of these results and brief description of each protein’s function was insufficient to fully justify our conclusions. We have now elaborated that while FADD was historically considered an adaptor of extrinsic apoptosis, it is now appreciated as a pleiotropic molecule with both anti- and pro-inflammatory signaling functions. FADD’s pro-inflammatory roles include inflammasome activation and activating NF- κ B through the FADDosome. We have now revised our section headings to avoid oversimplification.

(8) IFI16-PYD displays discontinuous behavior according to Figure S1H; however, it is not included in Figure 2D, but AIM 2 is.

We only tested a subset of FL proteins spanning different functions within diverse signalosomes. IFI16 was not included. Hence it could not be meaningfully included in Fig. 2D.

(9) To demonstrate that "Nucleation barriers facilitate signal amplification in human cells," constructs using APAF1 CARD, NLRC4 CARD, caspase-9 CARD, and a chimera of the latter are used to create what the authors refer to as apoptosomes. Even though puncta are observed, referring to these assemblies as apoptosomes seems somewhat misleading. In addition, it is not clear why the activity of caspase-9 was not measured directly, instead of that of caspase-3 and 7, which could be activated by other means.

We agree that describing our chimeric assemblies as “apoptosomes” could be misleading, and have now refrained from doing so. We measured caspase-3/7 instead of caspase-9 for purely technical reasons – we were unable to find any reliable caspase-9 activity assays that were also compatible with our optogenetic and imaging wavelengths. In any case, our data with the widely used caspase3/7 reporter dyes confirm comparably effective signal propagation from the CASP9 versions to their relevant endogenous substrate for apoptotic signaling (pro-caspase-3/7). The subsequent differences in cell death efficiency between the two versions of CASP9 (Fig. 3E) cannot be attributed to indirect effects of blue light stimulation, because both versions received the same treatment. Note our stated justification for using these DFDs in the HEK293T background is that these cells lack NLRC4 and CASP1 proteins and therefore the activity we measure is due to the direct optogenetic activation.

The polymerization of caspase-1 CARD with NLRC4 CARD, leading to irreversible puncta, could just mean that the polymers are more stable. In fact, not all DFDs form equally stable or identical complexes, which does not necessarily imply that a nucleation barrier facilitates signal amplification. Could this conclusion be an overstatement?

Figure 3C shows that the polymers don’t simply persist following the transient stimulus – they continue to grow. That is, the soluble protein continues to join the polymers for a net

increase even though there is no longer a stimulus directing them to do so. This means the drive to polymerize is independent of the stimulus, i.e. the protein is supersaturated. In the absence of supersaturation, a difference in stability would simply change the rates at which the polymers shrink. That we see continued growth instead of shrinkage therefore cannot be explained just by a difference in stability. Nevertheless, the reviewer's critique caused us to realize that increased persistence of the CASP1CARD polymers could contribute to signal amplification independently of supersaturation if they act catalytically (i.e. where each polymerized CASP9 subunit sequentially activates multiple CASP3/7 molecules), and we had not adequately considered this. Unfortunately, the relevant experimentalist has now moved on from the lab leaving us unable to conduct the necessary experiments to resolve these two effects in a timely fashion. Consequently, we have now tempered our interpretation of these data.

(10) To demonstrate that "Innate immune adaptors are endogenously supersaturated," it is stated on page 5 that ASC clusters continue to grow for the full duration of the time course and that AIM2-PYD stops growing after 5 min. The data shown in Figure 4F indicate that AIM2-PYD grows after 5 mins, although slowly, and ASC starts to slow down at ~ 13 min. Because ASC has two DFDs, assemblies can grow faster and become bigger. How is this related to supersaturation?

That AIM2-PYD assemblies appear to grow somewhat (although not significantly statistically) would be consistent with AIM2-PYD's sequestration into the growing ASC clusters. All that matters for our conclusion regarding ASC is that ASC assemblies grow following cessation of the stimulus, which we now describe quantitatively. Supersaturation is defined as the ratio of total concentration to saturating concentration, which is an equilibrium property. For a given protein concentration, the presence of two DFDs, each contributing their own interactions to overall stability of the assembly, will increase supersaturation relative to the individual DFDs. Importantly, growth will not occur if the protein concentration lies below its C_{sat} , no matter how many DFDs it has.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

It isn't clear what is implied by the final sentence of the Abstract. Some of the conclusions have a speculative tone and would be better described in less certain terms. The final sentence of the abstract should be omitted.

We have revised the abstract to add appropriate nuance but consider the final sentence to be both justified by our data and important to convey our findings to a broad audience.

How does the size and nature of the seed influence the outcome of these DFD interactions? Although some non-seeded experiments are described, the majority of the results are derived from seeded experiments. Further details about the seeds should be included. How is the size of the nucleus controlled, and will seeds of smaller or larger size generate the same pattern of results?

This is a very important question! The seeds comprised genetic fusions of each DFD to a condensate-forming domain, as described. While this system is insufficient to explore the size-dependence of nucleation, we are developing tools to do exactly that, for example our recently published multivalent nanobody against mEos3,[3] wherein we piloted its use to compare the size-dependence of ASC versus amyloid nucleation. Much further work will be needed to fully utilize this approach for the question of interest, and that is the subject of ongoing but open-ended work in the lab.

What is the implication of the observation that only ~20% of the DFDs exhibited a discontinuous transition from no to high AmFRET signal? Further discussion of the DFDs

that exhibit a continuous transition would enrich the manuscript.

We consider the relationship to mortality important for understanding this observation. In the discussion we now explain that each supersaturated protein in a death-inducing pathway imposes a risk of unintentional death. We speculate that evolution therefore minimizes the number of supersaturated DFDs by restricting them to central nodes in the network. That way, a small number of supersaturable DFDs can be continuously “repurposed” with new receptor proteins for each D/PAMP. Additionally, as stated in our response to the related critique, we felt it was important to focus this manuscript on the novel concept of functional supersaturation necessarily at the expense of signaling regulation through better understood mechanisms.

Were the initial experiments with DFDs unseeded (Figure S1, F-G)? Clarify this in the text. The morphologies of all the subcellular assemblies appear similar. It is not possible to distinguish between long filaments and spherical or amorphous puncta (Figure S1F-G). Higher magnification images that allow evaluation and comparison of morphology should be provided.

The initial experiments were unseeded, as now clarified in the legend. We believe there was a misinterpretation resulting from both panels (S1F and G) showing fibrillar examples. To clarify, we have now added panel S1H showing representative DFDs classified as “punctate”, which we hope the reviewer agrees are clearly distinct from fibrillar.

The ASC and CARD14 assemblies in Figure S1G show very distinct fibrillar structures emerging from the mNS-DFD seeds. Please provide further explanation of the nature of these. Do these resemble ASC and CARD assemblies generated as a result of native stimuli rather than mNS-DFD seeds?

The μ NS-DFD puncta contain numerous seeding competent sites, which presumably causes multiple fibrils to initiate and emanate from them. This and potential bundling of these fibrils produces the star-like shape. We have no reason to believe the internal structure of these fibers differs from native signalosome assemblies. For example, point mutations at native subunit interfaces that were previously shown to disrupt fibrilization and signaling likewise disrupt assembly in our DAmFRET experiments (Figure S2A). To our knowledge there exist no examples of high-resolution DFD fibril structures that were induced by native stimuli. However, recent work using super-resolution imaging confirmed that nigericin-triggered endogenous ASC specks comprise a network of filaments that superficially resembles our star-like assemblies.[4]

Figure S2B is presented as evidence that assembly is mediated by native-like interfaces rather than amyloid-like misfolding. These SDD-Age gels cannot be used to infer a native-like structure for the protein within the assemblies, only that the assemblies are (mostly) solubilised by incubation with sarkosyl. Many misfolding but non-amyloid-structure assemblies could be consistent with these results. Additionally, several of the samples appear to show insoluble aggregates within the wells, which could also be consistent with amyloid-type structures. What is the nature of these aggregates? Why is the NLRP3PYD sample so much more intense than the others? Why was FL-ZBP1 included when it does not contain a DFD? Why were no sarkosyl-resistant assemblies observed with RIPK3-RHIM when this is known to be highly amyloidogenic?

ZBP1 and RIPK3^{RHIM} were one of multiple proteins inadvertently included on the complete gel shown in the original figure that is not relevant to the manuscript; we have now spliced out these unnecessary lanes (indicated with dashed lines) to avoid confusion. We have found that the specific fragment of RIPK3^{RHIM} used in this experiment -- residues 446-464 -- does not allow for robust amyloid formation. We believe this is a steric artifact due to its small size (19 residues) relative to the fused mEos3, because a longer fragment (446-518) forms amyloid

robustly. However the latter construct was not available at the time this experiment was done. Nevertheless, another known amyloid protein, RIPK1^{RHIM}, does show the expected smears on this gel and suffices for the positive control for amyloid. We do not understand why the NLRP3^{PYD} sample is more intense than the others. However, this anomaly does not impact our conclusion that DFDs do not form sarkosyl-resistant smears that would be indicative of amyloid.

Expand on the concept of autoinhibited oligomerisation. Is this due to structural features? What might be the advantage of autoinhibited oligomerisation for these DFDs?

We have elaborated on this section in the results.

End of page 3, which "former set of adaptors" are referred to here? This is ambiguous.

We have replaced "former" with "innate immune".

Page 5, the authors state that a kinetic barrier governs the activity of inflammatory signalosomes. While under the circumstances generated in this particular system, there is a kinetic barrier to the formation of large fibrillar complexes, can the same be said to be true in cells that respond to signals? They experience a specific triggering event. This should be redrafted to distinguish between the specific trigger in cells (downstream of a binding-driven event) and the kinetic barrier to self-association observed in this model system.

Yes, our findings establish that a kinetic barrier governs signalosome activation. By engineering a triggering event that is more specific than natural triggering events (see Figure 3), we exclude the possibility that the cell first responds to the signal to create conditions that stabilize inflammasome formation. This means that regardless of what may happen with a natural trigger, the driving force for assembly clearly pre-exists and is therefore held in check by a kinetic barrier.

On page 6, the statement "...lifespan may be limited by the thermodynamic drive for inflammatory signal amplification" is not clear. While this is strictly true following the initial triggering event, isn't lifespan limited by the stochastic activation? These very general statements stray beyond what can be substantiated on the basis of the data presented here.

We believe the source of confusion here was our misuse of the term "lifespan". We have now replaced it with "life expectancy", which we believe is substantiated by our statements as written.

Overall, the work presents a compelling, comprehensive analysis of the seeded self-assembly of DFDs. It identifies distinct properties for assembly of these domains that may underlie their particular physiological roles. However, some of the statements are quite general and not substantiated.

Page 6. Is "end cell fate" the intended phrase?

We have revised the phrase.

The data regarding conservation of DFD-like modules and activity is interesting and probably deserves inclusion. However, without substantial evidence of expression levels (i.e., results) and a more complete understanding of these other systems, the statement "These results suggest that the function of DFDs as energy reservoirs preceded the evolution of animals" appears as an over-reach.

We demonstrated that sequence-encoded nucleation barriers of DFDs are shared across animal signalosomes (human, zebrafish, sponge). This is not trivial as such nucleation

barriers are uncommon even among targeted screens of prion-like proteins.⁵ Therefore, they appear to have existed in the basal animal. We have now omitted the data concerning bacterial DFDs as these systems are indeed much less understood, and the concerned pathways lack the tripartite architecture of animal signalosomes. We therefore revised the sentence in question by replacing “evolution” with “radiation”.

Only a small number of DFDs exhibit this behaviour, so why is the conclusion drawn that energy storage for on-demand signalling may be the principal ancestral function of DFDs?

The totality of the data supports this conclusion. Briefly (but elaborated in the text), 1) intrinsic nucleation barriers are unusual even among self-associating proteins, the vast majority of which (e.g. condensates) would suffice for the only other major function ascribed to DFDs -- bringing effectors close enough for proximity-dependent activation (which has been repeatedly demonstrated in DFD-replacement experiments), 2) nucleation barriers are nevertheless conserved in innate immune signaling pathway, 3) that they are limited to approximately one DFD in each pathway is consistent with evolutionary selection to minimize accidental death.

Are there any other adapters like MyD88 that are inconsistent with this hypothesis? Are any others known to be controlled by oligomer formation? How strong is the evidence for hexameric oligomers? If there is a threshold size for oligomers, how does this differ from a stable seed/nucleus that triggers assembly, as in the discontinuous transition?

These are all good questions related to critiques that we have now addressed.

The use of the term "privatisation" is likely not consistently understood across the community and should be explained. Is it simply meant to imply independent operation? How is it actually different from other forms of deployment of DFDs that exhibit continuous assembly? Are they not also independent? What is implied by the opposite of privatisation here? The term may introduce ambiguity in this context.

We have now omitted this term.

Is there strong evidence that well-validated physiologically relevant LLPS systems exhibit supersaturation at concentrations that are very different from those of the DFDs examined in this study?

No, and this is a major point. As discussed in the text (with references), LLPS is incompatible with cell-wide supersaturation to a comparable magnitude as crystalline transitions, which precludes them from driving signal *amplification*. This helps to explain why the active state of DFD assemblies is ordered, when it has been repeatedly demonstrated that signal *propagation* itself does not require ordering.

The paragraph discussing TIR domains and functional amyloids would be enhanced with a comparison of amyloid systems where seeded nucleation results in assembly of a polymer with significant conformational change in the constituent monomers.

We do not yet understand how DFDs (and TIR domains) in some cases exhibit amyloid-like nucleation barriers without overt conformational differences between monomers and polymers. Work is underway in the lab to test specific hypotheses, but such discussion would be too speculative for the present paper.

The statement "High specificity also insulates pathways from each other" should be elaborated to discuss the issue of highly similar monomers that apparently assemble into filamentous forms with minimal structural rearrangement. How is the specificity generated?

We have elaborated the paragraph.

The final paragraph is speculative and utilises language that detracts from the quality and rigour of the study. While important principles have been revealed, more discussion of the limitations of the work would allow readers to evaluate the significance of the study and could be used to effectively stimulate further efforts to study the multiple different mechanisms that underpin critical signalling pathways in innate immunity and control cell fate.

We have now revised the final paragraph and included an extensive discussion of the limitations of the work.

Reviewer #2 (Recommendations for the authors):

(1) For clarity, it would be useful to include the names of the proteins in the bottom table of STable1, and such information at the top and bottom tables can be connected.

We are unable to determine what is meant by this suggestion. Table S1 does not have a “top” and “bottom table”. Every entry in Table S1 and S2 contains the protein name, its most frequently used alias in the literature (when not the official name), and the corresponding Uniprot protein ID.

(2) The language used in the abstract makes analogies between scientific and mundane terms, which compromises clarity. For example, what is meant by the terms shown below?

(a) ".....specifically templated by other DFDs....."

We have revised this phrase.

(b) "...function like batteries, storing and converting energy for life-or-death decisions."

Batteries convert chemical energy into electrical energy or thermal energy. What is the electrical energy produced by DFDs? Is there any evidence that DFDs change the temperature of the cells or transfer heat?

We have now included a familiar example of a thermal battery that operates analogously to the manner we show for DFDs. As now elaborated extensively, such batteries operate via a physical rather than chemical process -- a change in the state of matter (solute to crystalline) of a supersaturated “phase change material” (this is an established term). This is exactly what we show is happening for DFDs. While it would be illustrative to measure the heat released upon DFD polymerization in cells, the much faster rate of heat transfer relative to molecular diffusion makes that impossible with present methods. Nevertheless, such measurements are unnecessary because disorder-to-order phase transitions are fundamentally exothermic.

(c) "....privatizing..."

We now avoid this term.

Using appropriate scientific terms to explain the scientific results presented in this manuscript will increase clarity. Analogously, it is difficult to understand what the title of the manuscript means, "Protein phase change batteries..."

We appreciate this critique and have removed “batteries” from the title to make the work more accessible to biologists. However, we reject the implication that such terminology is inappropriate. We presume the reviewer meant “unfamiliar” instead of “inappropriate”. The well-reasoned application of terms from other fields is standard practice and arguably essential to convey new concepts in biology. The modern biology lexicon is built on this. For

example, Robert Hooke co-opted “cell” from the architecture of monasteries. More recently cell biologists appropriated “condensates” from soft matter physics. In both cases, the term while initially foreign to biologists usefully introduced a concept that lacked recognized precedent in biology. Similarly, “phase change battery” provides an accurate analogy for the central finding of our work, and we have now elaborated this analogy in the text.

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<https://doi.org/10.7554/eLife.107962.2.sa0>